Orqix change record

Waves A to D of the Camunda gap plan: integration, AI and MCP, authoring, operations and analytics

Features implemented, enhanced and fixed while closing the Camunda gaps of doc 29, with everything needed to apply the same changes to another copy of the code.

Item	Value
Date	8 November 2026
Repository branch	`claude/camunda-features-gap-a441b7`
Base commit	`d5b0b774` (Merge pull request #22, develop)
Final commit of this record	`7e63a8ed`
Wave A tickets	CG-W01, W02, W03, W04, W08; CG-C01, C02, C03, C04, C05, C08 / CG-I04, C09, C10; CG-E10; ServiceNow ITSM package
Wave B tickets	CG-A01, A02, A03, A04, A05 (AI decisions); CG-M01, M02, M03, M05, M07 (agent tools); CG-M04 (Orqix MCP server)
Wave C tickets	CG-X01, F04 (expressions); CG-E01, E06 (loops, for each); CG-W05, W06, W07, E02, E08 (waits, schedules); CG-F01, F02, F05 (forms); CG-T04, T01, S03 (task actions)
Wave D tickets	CG-O01, O04, O05 (bulk, search); CG-O02, O03 (migration, moves); CG-O06, O07, P04 (alerts, retention); CG-N01, N03, N05 (analytics); completion: CG-D04, A06, A07, AI skills pages
Files changed	About 930 files across backend, compiler, contracts, frontend, SDK, examples and docs
Database migrations	V210–V242, V250, V260, V270–V275, V280, V290 (Waves A and B); V310–V401 (Waves C and D)
Verification	Compiler 1,224 tests, application 1,412 tests (0 failures, 5 opt-in skips), frontend 1,785 tests, typecheck, lint and API contract check: all pass
Status	Pushed to the branch; open as pull request #25 into develop (not merged yet)

1. What changed, in one page

Before this work Orqix could not connect to a typical SaaS API (for example ServiceNow) without writing an external worker, could not keep checking another system until a job was done, retried every failing call with one fixed delay, had no governed AI decision step, and could only call one MCP server per agent. Wave A fixes the integration side; Wave B adds AI decisions, agent tools and an Orqix MCP server.

Ticket	Feature	Type	Build note
CG-W01	Keep checking until done: a call step repeats a read-only check on a schedule until a condition says done, failed or the deadline passes	New	S129
CG-W03	Polling safety net for message waits: resume on a webhook or a slow check, whichever comes first	New	S129
CG-W08	Waiting for: one projection, panel and API saying what every request is waiting on; operator Check now / Extend deadline / Stop waiting; Insights 'Waiting on outside systems'	New	S129
CG-C01	Connection definition v1: path templates, query parameters with safe encoders, headers, DELETE, nested/form/multipart/binary bodies, row lists, pagination, error mapping, 120 s long calls	Enhanced	S130
CG-C02	Lookup reconciliation (idempotentByLookup) and idempotentNatural: no duplicate records after an uncertain create, no manual reconcile	New	S130
CG-W04	Per-step retry policy: exponential backoff with jitter, Retry-After for 429/503, circuit breaker and request limit per connection binding	Enhanced	S131
CG-C09	Connection health in Settings: success rate, latency, 429s, token failures, circuit state, notices	New	S131
CG-C08 / CG-I04	Secret providers: HashiCorp Vault, AWS Secrets Manager, Azure Key Vault, Google Secret Manager, with a secret-access audit that never records values	New	S132
CG-E10	Worker business errors: an external worker can report a business error that takes the step's Failed route or the section's error route	New	S132
CG-W02	Start and follow up: an API that answers 202 Accepted with a status location is followed by a Check-until step; status URLs must stay inside the binding	New	S133
CG-C03 / C04 / C10	Signed connector packages (.orqpkg) with review, approval by a second admin, install, upgrade, retire, Studio palette and an organization catalog	New	S133
CG-C05	Package triggers: verified webhooks (HMAC, JWT, basic) and polling triggers with watermark and dedupe	New	S133
ServiceNow	ServiceNow ITSM package: incidents, catalog orders, lookups, attachments, webhook and polling triggers, IT access request starter workflow	New	S133
CG-A04	Model provider SPI: OpenAI-compatible (OpenRouter, vLLM, Ollama), Anthropic, Bedrock, Azure OpenAI, Vertex/Gemini; residency rules; credentials from secret stores	New	S134
CG-A01 / A02 / A03 / A05	AI skills and the AI task step with guardrails, calibrated confidence, autonomy policy, evaluation gate, shadow mode, promotion, drift fallback, AI decision records and long-running AI tasks	New	S134
CG-M01 / M02 / M03	Agent tools: MCP tool servers with schema pins, operations, decisions, workflows and 'ask a person' as tools, durable agent loops with approvals and kill switches	New	S135
CG-M05 / M07	Opt-in redacted agent trace, agent dashboard, agent test suites with LLM-as-judge	New	S135
CG-M04	Orqix MCP server: assistants list, start, search, explain and decide under the user's own permissions, with confirmation for every write	New	S136
Waves C and D	Expressions, loops, waits and schedules, forms, task actions; bulk operations, migration and moves, alerts and retention, analytics, and the earlier AI and decision gaps (sections 11 and 12)	New	S137–S146
Integration fixes	Merge overlaps combined; seven defects found by the combined runs fixed (section 14)	Fixed	This record

2. How to apply these changes in another copy

Option A (recommended): bring the commits across with Git

If the office copy shares history with this repository (it contains commit `d5b0b774`), fetch the branch and merge it. Everything, including conflict resolutions and fixes, comes with it.

```
git fetch <this-repo> claude/camunda-features-gap-a441b7
git checkout -b wave-a <your-base-branch>
git merge --no-ff FETCH_HEAD
# or apply as patches:
git format-patch d5b0b774..HEAD -o orqix-patches
git am --3way orqix-patches/*.patch
```

Merging the branch is strongly recommended: Waves C and D were built on parallel branches and merged with many conflict resolutions (section 14), which a cherry-pick would have to redo by hand. For Waves A and B, when the office copy has drifted, cherry-pick in this order (the merges in between only combine branches; their resolutions are described in section 8 and must be redone by hand when cherry-picking):

#	Commit	Content
1	`b931c3a5`	docs: doc 29 gap analysis and plan, README link
2	`1411a52a`	compiler: Keep checking until done and message-wait safety net
3	`b19e7d09`	runtime: poll scheduler, safety-net checks, Waiting-for API
4	`c667c1c4`	studio: inspector, Waiting-for panel, simulator check sequences
5	`78865323`	integration: connection definition v1 and lookup reconciliation
6	`c07156bc`	step retry policy, Retry-After, circuit breaker, connection health
7	`003ea821`	secret providers with access audit, worker business errors
8	`9f80e10b`	fix: retryAfter in the paginated call result (merge fix)
9	`49541af6`	waits follow-ups after the merge (deferral, Retry-After, optional results, migration and repair coverage, purge, inspector, Insights, load test)
10	`1fc2d70b`	fix: poll purge migration renumbered V212 to V242 and made rerunnable
11	`4b744404`	docs: first (Wave A) change record
12	`0ad8a224`	Orqix MCP server (CG-M04, S136)
13	`515c5c4a`	connector packages, package triggers, start and follow up, ServiceNow package (S133)
14	`934618e8`	test: async variant in the call-and-wait inspector test
15	`e91666ea`	AI: model provider SPI and the aiTask step in the compiler
16	`04fa2cd7`	AI: skills, guardrail pipeline, async AI jobs, evaluations, shadow mode, decision records
17	`01ef7e1b`	AI: AiSkills API, evidence packs, tests, Studio model
18	`f8fea724`	AI: Studio inspector, AI decisions in run detail, mocks, contract fixes
19	`ad5067b0`	docs: S134 build note
20	`47290b66`	agent tools: tool servers, durable loops, trace, agent tests (S135)
21	`6b611f5b`	fix: merged grantable permissions (V290); mcpServer in the connection summary contract

Option B: re-implement by hand

Use sections 3 to 12 as the specification. Each section lists the contract fields, compiler rules, database migrations, API operations, permissions, configuration and the files that carry the change. The build notes `docs/build/S129` to `S146` in the repository hold the full detail and are part of the commits above.

Prerequisites and order

- JDK 21 (the build used `JAVA_HOME=C:/Users/karup/.jdks/jdk-21.0.12.1+1`); Maven wrapper `backend/mvnw` or the Gradle build; Node for the frontend.
- Database migrations run in version order: V210, V211, V220, V230, V240, V241, V242, V250, V260, V270–V275, V280, V290, then V310 to V401 in number order. They only add tables, columns, checks, functions and grants; no data is rewritten. V242 is safe on a database that already ran an earlier V212. V290 must come last: it rebuilds the grantable-permission checks that V250 and V260 both changed; V400 does the same after V350 and V370, and V401 extends the retention purge function.

- New grant-only permissions to hand out (none is in a role): `connector.manage`, `connector.approve`, `ai.approve-skill`, `ai.evaluate`, `run.bulk-operate`, `alert.manage`.
- New deployment settings: trusted package publisher keys (`orqix.connector-packages.trusted-keys.<keyId>`), AI model endpoints and profiles (`orqix.ai.model-providers.*`), the AI turn-state key secret (`orqix.ai.tasks.turn-state-key-ref`), and the MCP server settings (`orqix.mcp.*`) plus a Keycloak client `orqix-mcp-client` with an `orqix-mcp` audience mapper.
- Regenerate frontend API types after taking `contracts/openapi.yaml`: `npm run api:generate`, then `npm run api:check`.
- Compiler version moves to 0.3.0 (one bump for Wave A). Existing workflows compile to the same BPMN unless they use the new options.
- Every new external secret store is off by default; nothing changes until it is enabled (section 6).

3. Keep checking until done, safety net and Waiting for (S129)

What it does

A **Call a system** step on a read-only operation gets **Keep checking until…** (`config.poll`). Orqix calls the operation, maps the answer and continues on `next` when `until` holds, takes `failed` when `failWhen` holds, and takes `timedOut` at the deadline or when the checks are used up. The repetition lives inside the step, so the workflow graph stays acyclic.

```
"poll": {
  "every": "PT15M", "backoff": { "multiplier": 2, "maxInterval": "PT2H" }, "jitterPercent": 10,
  "until":    { "fieldRef": "/steps/checkIncident/state", "operator": "in", "value": ["6", "7"] },
  "failWhen": { "fieldRef": "/steps/checkIncident/state", "operator": "eq", "value": "8" },
  "giveUpAfter": "P5D", "maxChecks": 200,
  "onCheckError": { "maxConsecutive": 5, "then": "needsAttention" },
  "finalCheckAtDeadline": true, "showWhileWaiting": ["state"], "optionalResults": []
}
```

How long it waits. The first check runs immediately; each interval is the previous one times the multiplier, capped at `maxInterval`, with ±jitter. Example: every 5 minutes, multiplier 2, cap 1 hour, give up after 2 days gives 52 checks (0:00, 0:05, 0:15, 0:35, 1:15, then hourly, plus a final check at the deadline). The request resumes at the first check after the other system finishes, so the worst-case delay is the current interval.

A **Wait for a message** step can add a safety net (`config.fallbackCheck`): the same call settings with a slow interval. It resumes on the message or on the check, whichever comes first, and never twice.

Compiler rules

Code	Rule
WF_POLL_READONLY	The operation must be readOnly (also for the safety net)
WF_POLL_INTERVAL	every PT10S to P1D (safety net PT1M to P1D)
WF_POLL_BACKOFF	maxInterval at least every, at most P1D; multiplier 1 to 4
WF_POLL_DEADLINE	giveUpAfter PT1M to P90D and at least every
WF_POLL_TOO_MANY_CHECKS	Worst case at most 1,000 checks without maxChecks
WF_POLL_ROUTES	Exactly one timedOut path; a failed path when failWhen is set
WF_POLL_CONDITION / WF_POLL_OVERLAP	Conditions name mapped results; until and failWhen can never both hold
WF_POLL_SHOW / WF_POLL_OPTIONAL	Shown and optional fields must be kept results
WF_POLL_COMBINATION	No polling on an undo call or a call that raises an error on failure
WF_POLL_FALLBACK_MAPPING	Safety-net results map to declared message payload fields

Runtime design

- The step compiles to a triggerable service task. Entering it inserts a `poll_wait` row (V210). `PollScheduler` (tick 2 s, worker and all-in-one pods only) claims up to 20 due rows with `FOR UPDATE SKIP LOCKED`, a lease and a fence, runs the check through `EffectDispatcher` (egress, credentials, timeout, journal), and records the outcome only if it still holds the fence.
- One journal entry per check (occurrence = check number), so a pod that dies after the call is re-claimed and reuses the recorded answer instead of calling again.
- After downtime a due row gets exactly one check, never one per missed interval. Suspended requests are not checked; cancelling a request closes its rows in the same transaction.
- No payloads at rest: the row keeps status, an outcome code, a digest and up to three shown values.
- Deferred calls (open circuit or rate limit) move the next check to the end of the wait without counting a check or an error; `Retry-After` sets the next check, capped at the deadline and maxInterval.
- Release migration of a polling request only when the step settings are identical (`POLL_CHANGED`); the repair scan reports poll and wait-state drift; ended rows are purged after 30 days (V242).

API operations (tag Waits)

Operation	Endpoint	Permission
listRunWaits	GET /workspaces/{ws}/runs/{runId}/waits	run.read (initiator or participant)
searchWaits	GET /workspaces/{ws}/waits?kind=&connectionKey=&waitingLongerThan=	run.read
checkWaitNow	POST .../runs/{runId}/waits/{waitId}/check-now	run.operate, reason, audited
extendWaitDeadline	POST .../waits/{waitId}/extend-deadline	run.operate, reason, audited
stopWaiting	POST .../waits/{waitId}/stop	run.operate, reason, audited
getInsightsOutsideWaits	GET /workspaces/{ws}/insights/outside-waits	run.read

Database and configuration

Item	Detail
V210__poll_wait.sql	poll_wait table (state, checks, next check, deadline, lease, fence, digest, shown values)
V211__wait_state.sql	wait_state projection for every wait kind
V242__poll_wait_purge.sql	DELETE grant and ended_at index for the purge (renumbered from V212)
orqix.polling.tick	Scheduler tick, default 2 s
orqix.polling.retention / purge-delay	Ended-row retention, default 30 days; purge hourly

Main files: `PollTaskDelegate, PollScheduler, PollWaits, WaitStates, MessageWaits, RunRepair, ReleaseCompatibility, InsightsQueries` (runtime); `PollRules, PollSchedule, GraphRules, SimulationRun` (compiler); `PollFields.tsx, FallbackCheckFields.tsx, WaitingForPanel, OutsideWaits.tsx, pollSchedule.ts` (frontend); `contracts/workflow-v0.schema.json`.

4. Connection definition v1 and lookup reconciliation (S130)

A new sibling schema `contracts/connection-definition-v1.schema.json` (schemaVersion 1.0, kind http). Version 0.1 definitions are read, validated, compiled and executed exactly as before; the schema is chosen by schemaVersion. Workflows still map flat inputs; the wire shape is declared in the connection.

Addition	Contract	Safety rule
Path templates	/api/now/table/incident/{sysId} with typed pathParams (pattern, enum or const)	Percent-encoded; empty, '.', '..', '/', '\' and control characters refused; text needs a pattern or enum
Query	const, input, or template with encoder rfc3986 or servicenowEncodedQuery	Encoders are reviewed code; the ServiceNow encoder refuses ^ and line breaks in values
Headers	Static or mapped headers; responseHeaders readable	Authorization, Cookie, Host, Content-*, hop-by-hop, Proxy-*, Sec-*, X-Orqix-* refused
DELETE	method DELETE	Cannot be readOnly
Bodies	Templates; json, form, multipart, binary	Depth 5, 200 leaves; files only from this request's attachments, not quarantined, at most 10 MiB
Results	Array indexes, /[*] row lists, /@headers/Name	At most 100 items per list; depth 5
Errors	errors: [{status, code, bodyPointer?, message?}]	Mapped 4xx becomes a known rejection with failureCode; 401/403/408/429 keep platform meaning; 5xx cannot be mapped
Pagination	offset, cursor, pageToken, linkHeader with maxPages <= 10, maxItems <= 100	GET and readOnly only; next links must stay under the binding base URL
Timeouts	maxTimeoutSeconds up to 120 with long: true; workflow timeoutSeconds max raised to 120	Job lock re-extended when needed
Async, rate limit	Declared and validated	Execution comes with CG-W02 (next batch)

Lookup reconciliation (idempotentByLookup)

For systems without idempotency keys that store an external reference (ServiceNow `correlation_id`). Orqix writes a stable key (request reference, step and occurrence, for example `ORQ-1042-raise-1`) on every send. When the outcome is uncertain (timeout, reset, 408, 5xx, or a crashed attempt with an expired lease) it runs the read-only lookup: **found** records success with the found record and sends nothing again; **not found** allows a resend with the same key; a failed lookup leaves the outcome unknown as before. The lookup is retried 3 times over `lookupSettleSeconds` (default 30). `idempotentNatural` (PUT, PATCH, DELETE) retries like idempotentWithKey without a header; appending operations stay nonIdempotent.

Database: `V230__connection_v1_effect_classes.sql` widens the effect_journal effect-class check. Rule codes `CONN_V1_*` live in `ConnectionContracts.java`; `WF_MAPPING_RECONCILE_KEY` stops workflows mapping the key field.

Main files: `ConnectionContracts, V1Requests, V1Results, HttpExecutor, EffectDispatcher, EffectJournal, HttpTaskDelegate, RequestFiles, OpenApiConverter, ConnectionAdminService`; frontend `connectionSchema.ts, HttpInspector.tsx`; example `examples/connections/servicenow-table.connection.json` with WireMock fixtures.

5. Step retry policy, Retry-After, circuit breaker and connection health (S131)

```
"retry": { "maxAttempts": 6, "delay": "PT10S", "multiplier": 3, "maxDelay": "PT30M", "honorRetryAfter": true }
```

- Allowed on http, externalTask and sendAdapter steps. Without it a step keeps execution.maxAttempts and retryDelaySeconds and its BPMN is unchanged.
- Bounds: maxAttempts 1–10, multiplier 1–4, delay 5 s to 1 h, maxDelay at most 6 h. Rules `WF_RETRY_DELAY_INVALID, _TOO_SHORT, _TOO_LONG, WF_RETRY_MAX_DELAY_TOO_LONG, _BELOW_DELAY`.
- Attempt n waits min(maxDelay, delay × multiplier^(n–1)) ± 10 %, computed in `OrqixFailedJobCommandFactory` via `StepRetryPolicy`. Retry-After (seconds or HTTP date) of a 429/503 is used when it is not longer than maxDelay.
- Unknown and permanent outcomes still never retry automatically.
- **Circuit breaker per binding** (`connection_circuit`): 10 retryable failures in a row within 5 minutes open it; calls wait 1 minute (doubling to 15) **without using an attempt**; one trial call decides.

- **Request limit per binding**: `PUT /workspaces/{ws}/connections/{key}/rate-limit` (connection.manage), split across live pods.
- **Connection health**: one row per call kept 24 hours (no payloads); hour and day figures, last error, circuit state and notices CIRCUIT_OPEN, CIRCUIT_TRIAL, TOKEN_REFRESH_FAILING, LOW_SUCCESS_RATE, RATE_LIMITED_BY_SYSTEM, CALLS_DEFERRED in Settings → Connections.

Item	Detail
V220__connection_health_and_circuit.sql	connection_call, connection_circuit, connection_rate_limit, integration_pod; external_job retry columns
API	listConnectionHealth, getConnectionHealth (connection.read); putConnectionRateLimit (connection.manage)
Configuration	orqix.integration.circuit.failure-threshold (10), window (5m), cooldown (1m), max-cooldown (15m)
Behaviour change	A burst of 10 retryable failures across requests of one binding now pauses that binding for a minute instead of each request retrying on its own

6. Secret providers and worker business errors (S132)

Secret references

Reference	Store	Enable with
crm-token	Mounted file (Kubernetes/OpenShift Secret), as before	Always on
vault:<mount>/<path>#<key>	HashiCorp Vault KV v2 (token file or Kubernetes auth)	ORQIX_VAULT_ENABLED, ORQIX_VAULT_ADDR, ORQIX_VAULT_AUTH, ORQIX_VAULT_TOKEN_FILE or _K8S_ROLE
aws-sm:<id or ARN>#<key>	AWS Secrets Manager (EKS web identity or environment keys)	ORQIX_AWS_SECRETS_ENABLED, AWS_REGION
azure-kv:<vault>/<secret>[/<version>]	Azure Key Vault (workload or managed identity)	ORQIX_AZURE_KEYVAULT_ENABLED
gcp-sm:<project>/<secret>[/<version>]	Google Secret Manager (workload identity)	ORQIX_GCP_SECRETS_ENABLED

- SPI `com.orqix.integration.secrets.SecretProvider` behind the unchanged `SecretStore` methods. No cloud SDKs: small JDK HTTP clients (AWS Signature V4 included), so the dependency set does not grow.
- Remote values cached 30 s (`ORQIX_SECRETS_CACHE_TTL`, 0s disables); missing values and failures are never cached; store calls time out after 5 s.
- **Secret-access audit**: counts per workspace, reference, store and day (resolved, cache hits, not found, failed). `GET /workspaces/{ws}/secret-access?days=7` (connection.manage). No value, hash or length is ever stored or logged.
- Store references are accepted on connection bindings (and OAuth client-credential JSON secrets); adapters, triggers, Kafka and webhooks still use mounted files.
- Database: `V240__secret_providers_and_access_audit.sql` (secret_ref check, secret_access_count).

Worker business errors

- `POST .../external-jobs/{jobId}/fail` accepts `"errorCode": "SUPPLIER_BLOCKED"` (pattern ^[A-Z][A-Z0-9_]{1,39}$, final; retryable true or ORQIX_ codes give 422).
- The step takes its **Failed** route with outputs failureKind (workerError), failureCode, failureMessage, or with `raiseErrorOnFailure` the section's error route; otherwise the request needs attention as before.
- When the engine stops a worker step (section time limit, another error, request ended) its open job is cancelled and a late completion gets 409.
- Python SDK: `BusinessError(code, message)` or `Worker.fail_with_error(job, code, message)`. Studio: Failed handle and route on worker steps.
- Database: `V241__external_job_business_errors.sql` (external_job.error_code).

7. Connector packages, package triggers, start and follow up, ServiceNow (S133)

Start and follow up (CG-W02)

A v1 operation may declare `async`: accepted statuses (202), where the status location comes from (a header such as Location, or a body pointer), the status operation, poll interval and maximum wait. The location must have the binding's scheme, host, port and base path (`STATUS_URL_OUTSIDE_BINDING`) and match the status operation's path template (`STATUS_URL_UNEXPECTED`); its values become results `/@async/<name>`. In Studio, **Call and wait for completion** inserts a Check-until step on the status operation right after the call. A refused location is never followed.

Package format (CG-C03)

- Schema `contracts/connector-package-v1.schema.json`; a package is a ZIP (`.orqpkg`) with manifest, connections (v0 or v1, kind http), triggers, step templates, starter workflows, fixtures, docs, `checksums.sha256` and an Ed25519 JWS `signature.jws`.
- Import checks: at most 5 MiB packed, 10 MiB unpacked, 200 files of at most 1 MiB, compression ratio at most 100, safe paths only; every hash and the signature against trusted keys from deployment settings; every file against its schema; manifest counts equal the content; host patterns concrete; step templates and starter workflows compiled; fixtures replayed through the live response handling. Unsigned or untrusted packages need an audited override reason; a signature that does not match is always refused.

Lifecycle (CG-C04) and catalog (CG-C10)

Step	API	Permission and control
Import	POST /connector-packages (archive, unsignedReason)	fresh connector.manage; audited
Review	GET /connector-packages/{id}	Hosts, auth schemes, data leaving Orqix, operations by effect class, triggers, templates, fixtures, diff to the active version
Approve or reject	POST /connector-packages/{id}/decision (sha256, note)	grant-only connector.approve, never the importer; must name the exact hash reviewed
Install or upgrade	POST /connector-packages/{id}/install	Registers ordinary connection versions; never bindings or credentials; releases keep their pins
Bind and test	Existing binding and test endpoints	Base URL must match a package host (PKG_HOST)
Use	GET /connector-palette?q=	Studio lists operations and step templates; starter workflows via POST .../workflow-templates/{key}/workflows
Retire	POST /connector-packages/{id}/retire	Blocks new installs; pinned releases keep working
Catalog	GET /connector-catalog	Versions, signature status, installs, workflows pinning outdated versions

Package triggers (CG-C05)

- Webhooks at `POST /workspaces/{ws}/package-triggers/{id}/events` verified by the package's scheme: HMAC-SHA256 over timestamp and body (300 s tolerance), JWT (HS256 secret or RS256 public key; iss, aud, exp required, lifetime at most 300 s), or basic over HTTPS only (flagged weaker). Every failure is the same 401; repeats answer duplicate and never start twice.
- Polling triggers: one leased poller per trigger (`FOR UPDATE SKIP LOCKED`), watermark minus overlap, records handled in order, dedupe kept 7 days, circuit breaker and rate limit respected; watermark formats iso8601 and servicenowDateTime.
- At most 10 per workspace; start or message as the person who set the trigger up.

ServiceNow ITSM package

Source `examples/packages/servicenow-itsm/`, signed build `examples/packages/servicenow-itsm-1.0.0.orqpkg`. Connections `servicenow-itsm` (createIncident by lookup on correlation_id, getIncident, findIncidentByCorrelation, lookupUser, listChangedIncidents, updateIncidentFields

natural, addWorkNote and attachFile non-idempotent) and `servicenow-catalog` (orderCatalogItem by lookup, findRequestByReference, getRequestItem). Triggers incident-updated (signed webhook) and incident-changes (poll every 2 minutes). Starter workflow **IT access request**: raise the incident, wait for the webhook with a safety-net check every 2 hours for up to 5 days, then confirm the outcome. The README contains the ServiceNow business rule script; check it against your ServiceNow release.

Database `V250__connector_packages.sql` (connector_package_version, _install, _connection, package_trigger, package_trigger_record). Settings: `orqix.connector-packages.trusted-keys.<keyId>`, `development-hosts`, `poll-tick` (2 s), `poll-lease` (60 s), `record-retention` (7 days). Main files: `AsyncStatus, PackageArchive, PackageValidator, ConnectorPackageService, PackageTriggerService, PackagePolls, ConnectorPackageController`; frontend `ConnectorPackagesPanel.tsx, packageSteps.ts, Palette.tsx`.

8. AI decisions: providers, skills, the AI task step, evaluation (S134)

Model provider SPI (CG-A04)

- `com.orqix.ai.provider.ModelProvider` with JDK HTTP adapters (no SDKs): OpenAI-compatible (OpenRouter, vLLM, Ollama, LM Studio; strict JSON schema), Anthropic Messages (prompt caching, strict output), AWS Bedrock Converse (Signature V4 or API key), Azure OpenAI deployments, Google Vertex and the Gemini API.
- `ProviderRegistry` applies the existing profile policy (excluded models, allowlist, price caps, budgets) and a per-workspace residency rule; credentials resolve through the S132 secret providers.
- Settings under `orqix.ai.model-providers`: endpoints (kind, base URL, secret ref, region, project, location, api-version, deployments, residency tags, strict schema, logprobs, timeout) and profiles (provider, model, prices).

AI skills and the AI task step (CG-A01)

- Skills are shared, versioned assets (`contracts/ai-skill-v1.schema.json`): mode (classify, extract, score, recommend, summarize), instructions, typed inputs with data classification, labels with risk and adverse flags, examples, model profile, confidence method, evaluation targets, limits, promotion and drift settings. A different person holding `ai.approve-skill` approves a version; approved versions never change.
- New workflow step `aiTask`: skill, mapped inputs, autonomy (shadow, suggest, autoWithReview, auto; default suggest), minConfidence (0.8), review sample percent and per-input redaction. Routes: one per label, `lowConfidence`, `needsReview`, `aiUnavailable`.
- Compiler rules `WF_AI_*` include the publish gate `WF_AI_EVALUATION_REQUIRED` (a passing evaluation on the pinned profile within 30 days; shadow mode exempt) and `WF_AI_AUTONOMY_NOT_PROMOTED`.

Guardrail pipeline

#	Stage	What happens
1	Admission	Approved pinned version, residency, evaluation, budget; effective autonomy capped by policy
2	Minimization	Only mapped values; classification and redaction (mask, hash, drop); regulated workspaces never send sensitive fields
3	Prompt	Pinned instructions and examples; request values wrapped as data, never instructions
4	Provider call	Through the SPI with deadline (at most 10 minutes) and cost ceiling
5	Validation	Schema check with one repair turn
6	Consistency	Optional N samples, majority and agreement
7	Calibration	Raw score mapped to observed accuracy from the evaluation run
8	Routing	Shadow continues; suggest goes to a person; regulated adverse labels always go to a person; auto only for policy-listed low-risk labels
9	Record	AI decision record written with the step completion

Evaluation, shadow mode, records and long-running tasks (CG-A02, A03, A05)

- Evaluation suites and runs (`ai.evaluate`): accuracy, per-label precision and recall, confusion matrix, cost, p95 latency, calibration buckets, threshold trade-off. Shadow dashboard; one-step promotions approved by a second person; weekly drift check with automatic fallback to suggest.
- Human overrides become evaluation candidates; a review queue lists sampled automatic decisions.
- Workspace AI policy `GET/PUT /ai-policy`: regulated flag, residency, input retention (digest by default), retention days, low-risk labels, classification actions.
- AI decision record per task (`ai_decision_record`): skill, model, prompt hash, input digest, output, confidence, route, guardrails, human outcome, cost; shown in run detail and run evidence packs (no values).
- AI tasks run as queued jobs (`ai_task_job`) on worker pods with leases, encrypted turn state (AES-256-GCM) so a crash resumes from the last completed turn; deadline up to 10 minutes.

Database `V260__ai_skills_and_decisions.sql` (ai_skill, ai_skill_version, ai_skill_promotion, ai_eval_suite, ai_eval_run, ai_workspace_policy, ai_task_job, ai_turn_state, ai_decision_record, ai_eval_candidate; ai_generation kinds). 27 new API operations (tag AiSkills). Settings `orqix.ai.tasks.*` (lease 60 s, 4 threads, 3 attempts, turn-state key). Main files: `ProviderRegistry, AiSkills, AiTaskEngine, AiTaskDelegate, AiTaskWorker, TurnVault, AiTaskRules, AiSkillContracts`; frontend

`AiTaskInspector.tsx`, `AiDecisionsPanel.tsx`.

9. Agent tools, durable agent loops, trace and tests (S135)

- **Tool servers (CG-M01):** connection kind `mcpServer` holds the tool catalog with administrator-written descriptions and allow-listed resources. `POST .../connections/{key}/tool-snapshot` pins each tool's input schema by SHA-256; a changed or missing tool is refused (`refusedSchemaChanged`), recorded and notified. Agents name tools `server.tool` so two servers cannot shadow each other. Sampling is always refused.
- **Orqix things as tools (CG-M02):** connection operations (effect class and journal inherited; non-idempotent only under an approved write tool), pinned shared decision tables, pinned child workflows and 'ask a person'.
- **Durable loops (CG-M03):** an agent operation with `loop` runs one model turn per job execution and waits between turns without using attempts; limits maxTurns at most 30, maxToolCalls at most 50, maxDuration up to P7D, budget per run; approval before chosen tools; kill switches per agent and server; encrypted turn state cleared when the loop ends. Human requests appear on the new **Agents** page.
- **Trace and dashboard (CG-M05):** off by default; when switched on, inputs, turns and tool calls are kept 1–30 days (default 7) after redaction of e-mails, card-like numbers, tokens and named fields.
- **Agent tests (CG-M07):** suites with mock tool fixtures, tool-sequence assertions and LLM-as-judge assertions with a pinned judge model.
- No new permission: pins and switches need connection.manage, the trace setting workspace.manage-access, trace and dashboard run.operate.

Database V270 (mcp_tool_snapshot, mcp_tool_drift), V271 (agent_kill_switch), V272 (agent_loop, agent_human_request with a separation-of-duties check), V273 (agent_trace_setting, agent_trace_event), V274 (agent_test_suite), V275 (notification kind agentToolChanged). 13 new API operations (tag AgentTools). Main files: `McpClient, AgentExecutor, AgentLoops, AgentLoopScheduler, AgentHumanRequests, McpToolPins, AgentToolRules, AgentTrace, AgentTests`; frontend Agents page.

Security checklist covered by tests

Threat	Control
Tool poisoning	Administrator-written descriptions only
Rug pull	Schema hash pins and list_changed re-checks
Shadowing	server.tool namespacing, unique model names
Prompt injection	Tool output and resources treated as data; schema-valid results only
Token passthrough	Per-binding or delegated credentials; Orqix tokens never forwarded
Exfiltration through arguments	Data classes per tool; refused values never sent
Runaway cost or loops	Turn, call, time and budget limits; kill switches
Unapproved writes	Approved write tools and approval-before-tool with separation of duties
Cross-tenant access	Everything workspace-scoped; foreign principals get 404

10. Orqix MCP server for AI assistants (S136)

- Endpoint `/mcp` (Streamable HTTP, protocol 2025-11-25). Off by default per workspace; switched in Settings → AI assistants (`workspace.manage-access`).
- Authorization per the MCP specification: Orqix is an OAuth protected resource (RFC 9728 metadata at `/.well-known/oauth-protected-resource/mcp`); tokens must be audience-bound to the MCP resource; REST tokens are refused and MCP tokens are refused by the REST API; nothing is forwarded.
- Tools: list_workflows, search_requests, get_request, explain_request, list_my_tasks, get_task, decide_task, send_message, confirm and a generated start_<workflow> for workflows marked `exposeAsTool`. Results are the same as the REST API for that person, field by field.
- Every write is prepared, shown and confirmed: a summary and a 5-minute confirmation id; confirm runs it once with idempotency; MCP elicitation is used when the client supports it.
- Audit `mcp.tool-called` for every call without values; 60 calls a minute per person per pod.

Database `V280__mcp_server.sql` (mcp_workspace_setting, mcp_confirmation). API getMcpServer, setMcpServer, listMcpCalls. Keycloak dev realm client `orqix-mcp-client` (PKCE). New module `com.orqix.mcp`; frontend `McpServerPanel` and the Studio flag 'Offer to AI assistants'.

11. Wave C: authoring power, forms and task actions (S137 to S141)

Expressions, Calculate step and computed form fields (S137: CG-X01, CG-F04)

- A total, side-effect-free expression language (`com.orqix.compiler.expr`, `frontend/src/lib/expr`): decimal arithmetic (34 digits, half-even, declared rounding), text, date and working-day functions on the S27 calendars, list functions (count, sum, average, min, max, filter, any, all), if-then-else, in, three-valued and/or, coalesce, isEmpty. No loops, recursion or I/O. Limits: 2,000 characters, 300 nodes, depth 30, 100,000 evaluation steps.
- Java and TypeScript engines verified on one shared file of 203 cases (`examples/fixtures/expression-cases.json`).
- New **Calculate** step (named outputs from formulas) with a guided editor; formulas in conditions and in input mappings (= prefix); computed form fields and cross-field rules evaluated the same way in the browser and on the server; `evaluateExpression` API.
- Merge decision: time-of-day and date-time fields (S140) are not available in formulas, like files, because comparing them as text is wrong across time-zone offsets.

Bounded loops and for each on any step (S138: CG-E01, CG-E06)

- A connection back to an earlier step with `maxTraversals` 1–10 and a required `limitReached` route; the compiler proves every cycle is bounded. Tasks, effects, SLAs and decision records carry an occurrence number; separation of duties and effect safety hold on every pass.
- `forEach` (sequential or parallel, completion condition, collected outputs) on calls, external workers and AI tasks, up to 500 items with per-item effect keys; request lists can hold up to 500 items.
- Migration V310. Release migration refuses requests inside a loop unless the loop is identical.

Working-time waits, reminders, wait for the first, message handlers, scheduled starts (S139)

- Working calendars as shared, versioned assets; waits and deadlines in working time (`workingDuration`, `calendarRef`).
- Reminders while waiting (`whileWaiting.remind`: every, maxTimes, adapter, template), journaled with stable message ids; a recipient from a request field only if its domain is approved by the adapter at send time.
- `waitForFirst` step with 2–5 branches (message, signal, callback, time, check), compiled to an event-based gateway; message handlers (event subprocess on a request message, interrupting or alongside).
- Scheduled starts: cron or RRULE in a time zone, optional working calendar, catch-up and overlap policies, a leader-elected scheduler. Migrations V320–V322.

Field types, lookup fields and guest response forms (S140: CG-F01, CG-F02, CG-F05)

- Time and date-time (with zone), checklist and tag-list fields in the schema, both validators, the renderer and Studio.
- Lookup fields: options from a pinned read-only connection operation (for example ServiceNow users), search-as-you-type through `searchLookupOptions` under the requester's permissions, with rate limits and server-side re-validation of the chosen value.
- Guest response forms: a one-time, expiring, signed link lets an outside person answer a wait with typed values and files; approved domains, rate limits, revocation, audit; page at `/guest/:workspaceId`. Migration V330.

Escalation, actionable tasks, step templates (S141: CG-T04, CG-T01, CG-S03)

- Escalation policies on approvals: up to five levels (notify a supervisor group, reassign, alert operators) after a number of hours or working hours; each level fires once and only while the task is open.
- Actionable tasks by email, Teams and Slack: claim, approve or reject with confirmation; signed, expiring links bound to the task and person (`/act/:token`) or linked chat identities; same checks as the task API. Verified against stub transports only.
- Workspace step templates: save a configured step, versioned, searchable in the palette. Migrations V340–V342 (V340 adds notification kind taskEscalated).

12. Wave D: operations, analytics and earlier gaps (S142 to S146)

Bulk operations, search by data, job dashboard (S142: CG-O01, CG-O04, CG-O05)

- Bulk retry, cancel, suspend, resume or move-to-release by filter: a dry run with the exact count, execution only after that count is confirmed with a reason, a hard cap (1,000, at most 10,000), background batches with per-item results, pause/resume/cancel, rate limit and audit. A filter can never act on more requests than the dry run showed (the Camunda 8.10 lesson). New grant-only permission `run.bulk-operate`.
- Searchable request fields (`searchable: true`, up to 10 per workflow) indexed at start and on corrections; run filters by those fields and by attention or error code, permission-filtered like run reads.
- Operations dashboard: external job queues by topic, workers last seen, connection call volumes and errors, poll checks and package trigger lag. Settings `orqix.bulk.*`. Migrations V350–V353.

Instance migration with a step mapping, guarded moves (S143: CG-O02, CG-O03)

- Move a suspended request to another release with explicit step mapping, a plan that shows every refusal reason first, and batches of up to 100 requests of the same release.
- Guarded move: skip a step, run it again or move to another step; recorded calls are never sent twice; a second operator approves in production or regulated workspaces, within 24 hours. Migration V360.

Alert rules, retention and archiving, history purge (S144: CG-O06, CG-O07, CG-P04)

- Alert rules on signals (requests needing attention, SLA breaches, open circuits, token failures, dead letters, failing checks, AI drift) with thresholds, windows, quiet periods and resolve notices, delivered through adapters or webhooks (stubs in tests). New grant-only permission `alert.manage`; settings `orqix.alerts.*`.
- Per-workflow retention: ended requests are archived (evidence pack plus export data) to the file system or an S3-compatible store, then purged by `orqix_purge_run`; legal holds per request or workflow block purges; engine history is purged in batches. Settings `orqix.retention.*`. Migrations V370–V371.

Analytics read model, heatmaps, KPI targets, human work (S145: CG-N01, CG-N03, CG-N05)

- Reporting tables fed incrementally from the run timeline through an outbox (restart-safe, idempotent, with a backfill); heavy reports never read the operational tables.
- Heatmaps on the Studio canvas and run diagram (counts, average and p95 per step, path frequency); per-workflow KPI targets with a business-value view; waiting versus working time, load and SLA attainment per group. Migration V380.

Completion of earlier gaps (S146)

- Formulas as decision-table rule tests and results (CG-D04), with Java/TypeScript parity, evidence and refusal where DMN or CSV cannot represent them.
- Document extraction (CG-A06): an AI `extract` mode over attachments with document-type templates, OCR providers (local Tesseract, Azure Document Intelligence and AWS Textract adapters verified against stand-ins) and a verification screen beside the document.
- Operator 'Explain this request' (CG-A07): read-only, from the timeline, waits and compiled model; never acts.
- Studio pages for AI skills: authoring, approval, evaluations, shadow dashboard, promotions, drift, review queue and AI policy. Migrations V390 and V394.

Wave C and D migrations: V310, V320–V322, V330, V340–V342, V350–V353, V360, V370–V371, V380, V390, V394, plus the merge migrations V400 and V401. 95 new API operations. New grant-only permissions run.bulk-operate and alert.manage.

13. Plan document

`docs/29-camunda-gap-implementation-plan.md` compares Orqix with Camunda 7 and 8 up to 8.10, lists 16 Camunda drawbacks with the design rule each teaches, specifies waiting patterns, connector packages, AI decisions and MCP tools in detail, and contains the full gap matrix and delivery waves. The README reading path links it.

14. Fixes made while integrating

Each feature was built in parallel on its own branch and merged here. These are the problems the combined builds found and how each was resolved. Redo them by hand only when cherry-picking feature commits without the merge commits.

#	Problem	Fix	Where
F1	Retry and connection v1 both changed the call result record (`HttpExecutor.CallResult`) and the failure record (`EffectDispatcher.Failed`)	Records carry all fields (page, failureMessage, retryAfter / failureCode, failureMessage, retryAfter) with compatible constructors; the circuit guard wraps the v1 call path (lookups and pagination included)	Merge `22c7c59b`
F2	Paginated call result built with the old field count; the app did not compile	Pass the new retryAfter field (null) in the paginated result	`9f80e10b`
F3	Poll checks did not handle a deferral from the circuit breaker or rate limit, so a held-back check counted as an error	Scheduler catches the deferral, moves the next check without counting a check or error, gives up at the deadline; Retry-After used for the next check	`49541af6`
F4	Migration V212 sorted below V220–V241; a database at V241 would refuse it	Renumbered to V242 and index created IF NOT EXISTS, safe on a database that ran V212	`1fc2d70b`
F5	Worker business errors and step retry both changed the external-job failure branch	Business error routing first, then retry with the step's computed delay	Merge `b6adb99f`
F6	**Permissions lost:** V250 (packages) and V260 (AI) each rebuilt the grantable-permission database checks with their own list, so after V260 connector.manage and connector.approve could no longer be granted	New migration V290 rebuilds both checks with all twelve grantable permissions; a new test grants and revokes each one	`6b611f5b`
F7	The connection list contract did not allow kind mcpServer, so listConnections broke its OpenAPI contract as soon as a tool server existed (found by the secret-isolation test's contract check)	Added mcpServer to the ConnectionSummary kind enum; frontend types regenerated	`6b611f5b`
F8	Permissions, access admin, OpenAPI enums, the compiler's packaged schemas and the access panel were extended by both the packages and AI branches	Combined both lists everywhere (connector.* and ai.*; both new schemas packaged)	Merge `edd6d73f`
F9	Independent additions in the same places (simulator, test stubs, route rules, BPMN attributes, mock handler lists, security-matrix tables, connection admin constructor, admin guide)	Kept both sides; for example worker steps get the Failed route next to polling's timedOut, the stub serves both /reports/ and /mcp/ paths	Merges
F10	Generated frontend API types conflicted	Regenerated from the merged OpenAPI and checked	Merges
F11	CI only: an undefined permission label on two AI operations, and seven inline OpenAPI descriptions with commas parsed as bogus fields (Redocly errors)	Existing taskParticipant label; descriptions quoted	`e680c28a`
F12	Connector package archives depended on the build machine's time zone (ZIP entry times), so CI's byte check failed	Fixed local entry time (setTimeLocal); package rebuilt, signature unchanged	`7336909c`
F13	list_workflows silently stopped at 200; package tests left workspace-wide grants behind	truncated flag and note; package tests revoke the grants they create	`ca847361`
F14	Browser tests: a stale expected message and a typing race in the assistant test	Updated message; wait for each reply	`29b0a045`

#	Problem	Fix	Where
F15	Wave C: expressions and loops both rebuilt flow conditions in BPMN; waits and loops both extended the transition record, enums, schema and Studio kinds	Combined: loop structure plus expression conditions; the transition carries maxTraversals and branch	Merges
F16	Wave C: the forms branch added time and date-time field kinds unknown to the expression rules (no compile)	Formulas treat them as unavailable, like files	`245d172c`
F17	Wave C: step-kind count test, reminder code using a compiler internal (architecture rule), two more comma descriptions, and a palette whose 19 buttons shrank below the 24-pixel touch size	Count 22; shared JSON helper; quoted; buttons keep their size and the rail scrolls	`0f3f995d`, `9ef01db3`, `4aa6eadb`, `697ba95c`
F18	**Permissions lost again:** V350 and V370 each rebuilt the grantable-permission checks, so after V370 run.bulk-operate could not be granted	V400 rebuilds both checks with all fourteen permissions; the grant-everything test covers it	Merge
F19	**Retention purge incomplete:** the purge function did not know ten request-linked tables added by other Wave D slices (moves, bulk items, search values, analytics, document verifications)	V401 extends it; analytics facts are deleted with the request (second-model review)	`fa0319cd`
F20	A test from the completion work built the attention record without the new errorCode field	Passes null for errorCode	`86bd0d69`
F21	CI only: the case-stage test assumed exactly two operators; S143 added a second operator test user, so three people were (correctly) told	The test counts recipients and checks each is told once	`7e63a8ed`

15. Verification (final combined run)

Suite	Result
Compiler (backend/compiler)	1,224 tests, 0 failures
Application (backend/app, PostgreSQL and Flowable via Testcontainers, WireMock, Vault container, stub MCP)	1,412 tests, 0 failures, 5 skipped (live-AI tests, the opt-in polling load test and existing opt-ins)
Polling load test (run by the waits session)	2,000 concurrent polling waits against WireMock; measured check rate recorded in S129
Frontend (Vitest, 239 files)	1,785 tests pass; typecheck, ESLint and OpenAPI type check pass
Python worker SDK	38 tests pass (run by the secrets session)
Not run	Docker stack, real-browser journeys, live ServiceNow, live AI providers or cloud secret stores, production gates

16. Known limits and what comes next

- Each poll check adds one effect-journal row (up to 1,000 per step occurrence).
- Rate limits (bindings, MCP calls) are per pod; exact global limits need a shared limiter (decision D-CG8).
- No Studio editor for `config.retry` yet; it is set in the workflow JSON or imported BPMN.
- Connections: no XML/SOAP templates, odataFilter or jql encoders, Basic/JWT/mTLS bindings, or row lists in the OpenAPI import yet. Packages: no code connectors (CG-C06) or generators (CG-C07); fixture conformance against a sandbox instance not run; ServiceNow verified against a ServiceNow-shaped stub only.
- AI: agents still use their own OpenRouter client rather than the provider SPI; AI decision records past their retention date are not purged on their own yet; every provider and OCR adapter is verified against stand-ins only. (Document

extraction, operator explain and the AI skills pages were completed in Wave D, section 12.)

- Agent tools: no MCP prompts or resource templates; no compiler warning for unpinned tools; agent waits not yet in the Waiting-for panel.
- MCP server: no resources or prompts; no file upload; no CORS for browser-hosted assistants.
- Waves C and D: live email, Teams, Slack, PagerDuty, OCR and model accounts are verified against stand-ins only; for each on wait steps, loops around waits and child steps, renaming waits during migration and moves inside parallel sections are not built; no report builder, outlier analysis or event-log export yet; time fields cannot be used in formulas; the analytics projector has no volume measurement yet.
- Next (doc 29 Wave E): Git sync, release bundles, test-as-code CLI, SDKs, OpenTelemetry tracing, scale qualification, resource-level permissions and field protection; then the report builder and remaining items.

Appendix: files changed per commit

A = added, M = modified, D = deleted, R = renamed. Generated from Git.

b931c3a5 docs: Camunda gap analysis, drawbacks and implementation plan

```
M README.md
A docs/29-camunda-gap-implementation-plan.md
```

1411a52a feat(compiler): Keep checking until done and message-wait safety net (CG-W01, CG-W03)

```
A backend/compiler/src/main/java/com/orqix/compiler/PollSchedule.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FailureRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/PollRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/PollRulesTest.java
M contracts/workflow-v0.schema.json
A examples/connections/order-status.connection.json
```

b19e7d09 feat(runtime): poll scheduler, safety-net checks and the Waiting for API (CG-W01, CG-W03, CG-W08)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/runtime/api/WaitDtos.java
A backend/app/src/main/java/com/orqix/runtime/api/WaitsController.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixRoute.java
A backend/app/src/main/java/com/orqix/runtime/engine/PollTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/RuntimeProjectionListener.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageDelivery.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/OperatorService.java
A backend/app/src/main/java/com/orqix/runtime/internal/PollConditions.java
A backend/app/src/main/java/com/orqix/runtime/internal/PollScheduler.java
A backend/app/src/main/java/com/orqix/runtime/internal/PollWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunStore.java
A backend/app/src/main/java/com/orqix/runtime/internal/WaitStates.java
A backend/app/src/main/java/com/orqix/runtime/internal/WaitsService.java
A backend/app/src/main/resources/db/migration/V210__poll_wait.sql
A backend/app/src/main/resources/db/migration/V211__wait_state.sql
A backend/app/src/test/java/com/orqix/runtime/PollWaitsTest.java
M backend/app/src/test/java/com/orqix/support/Stub.java
A backend/app/src/test/java/com/orqix/support/TrackingStub.java
M contracts/openapi.yaml
```

c667c1c4 feat(studio): Keep checking until inspector, Waiting for panel, simulator check sequences; S129 note

```
M contracts/openapi.yaml
M docs/build/README.md
A docs/build/S129-check-until-and-waits.md
M frontend/src/api/schema.d.ts
M frontend/src/features/runs/RunDetailPage.tsx
A frontend/src/features/runs/WaitingForPanel.test.tsx
A frontend/src/features/runs/WaitingForPanel.tsx
M frontend/src/features/studio/inspector/HttpInspector.tsx
A frontend/src/features/studio/inspector/PollFields.test.tsx
A frontend/src/features/studio/inspector/PollFields.tsx
A frontend/src/features/studio/model/pollSchedule.test.ts
A frontend/src/features/studio/model/pollSchedule.ts
M frontend/src/features/studio/test/StepSettings.tsx
M frontend/src/features/studio/test/scenarioModel.ts
M frontend/src/mocks/callbacks.ts
A frontend/src/mocks/pollSimulator.test.ts
M frontend/src/mocks/simulator.ts
A frontend/src/mocks/waits.ts
```

78865323 feat(integration): connection definition v1 and lookup reconciliation (CG-C01, CG-C02)

```
M backend/app/src/main/java/com/orqix/catalog/export/WorkspaceExportService.java
M backend/app/src/main/java/com/orqix/integration/ConnectionDefinition.java
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
M backend/app/src/main/java/com/orqix/integration/EffectJournal.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
A backend/app/src/main/java/com/orqix/integration/RequestFiles.java
A backend/app/src/main/java/com/orqix/integration/V1Requests.java
A backend/app/src/main/java/com/orqix/integration/V1Results.java
M backend/app/src/main/java/com/orqix/integration/internal/ConnectionAdminService.java
M backend/app/src/main/java/com/orqix/integration/openapi/OpenApiConverter.java
M backend/app/src/main/java/com/orqix/integration/openapi/OpenApiReview.java
M backend/app/src/main/java/com/orqix/integration/package-info.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
A backend/app/src/main/java/com/orqix/runtime/internal/RunRequestFiles.java
A backend/app/src/main/resources/db/migration/V230__connection_v1_effect_classes.sql
A backend/app/src/test/java/com/orqix/api/ConnectionV1ApiTest.java
M backend/app/src/test/java/com/orqix/api/WorkspaceExitExportApiTest.java
A backend/app/src/test/java/com/orqix/integration/ServiceNowFixturesTest.java
A backend/app/src/test/java/com/orqix/integration/V1RequestBodiesTest.java
M backend/app/src/test/java/com/orqix/integration/openapi/OpenApiConverterTest.java
A backend/app/src/test/java/com/orqix/runtime/ServiceNowConnectionTest.java
M backend/app/src/test/java/com/orqix/support/OpenApiContract.java
A backend/app/src/test/java/com/orqix/support/ServiceNowStub.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M backend/compiler/build.gradle
M backend/compiler/pom.xml
A backend/compiler/src/main/java/com/orqix/compiler/ConnectionContracts.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Connections.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FailureRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/schema/SchemaValidation.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/ConnectionV1CasesTest.java
A contracts/connection-definition-v1.schema.json
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
A docs/build/S130-connection-definition-v1.md
M docs/build/S4-workflow-compiler.md
A examples/connections/servicenow-table.connection.json
A examples/fixtures/connection-v1-cases.json
A examples/fixtures/servicenow-table.wiremock.json
M frontend/src/api/schema.d.ts
M frontend/src/features/runs/operatorText.ts
M frontend/src/features/studio/connectionSchema.ts
M frontend/src/features/studio/inspector/HttpInspector.tsx
A frontend/src/features/studio/inspector/HttpInspector.v1.test.tsx
M frontend/src/features/studio/useConnectionLookup.ts
M tools/contract-check/contract_check.py
```

c07156bc feat: step retry policy, Retry-After, circuit breaker and connection health (CG-W04, CG-C09)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/integration/CallDeferredException.java
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
A backend/app/src/main/java/com/orqix/integration/RetryAfter.java
A backend/app/src/main/java/com/orqix/integration/api/ConnectionHealthController.java
A backend/app/src/main/java/com/orqix/integration/health/BindingRateLimiter.java
A backend/app/src/main/java/com/orqix/integration/health/CallGuard.java
A backend/app/src/main/java/com/orqix/integration/health/ConnectionCircuit.java
A backend/app/src/main/java/com/orqix/integration/health/ConnectionHealth.java
A backend/app/src/main/java/com/orqix/integration/health/ConnectionHealthService.java
M backend/app/src/main/java/com/orqix/runtime/engine/EngineCommands.java
M backend/app/src/main/java/com/orqix/runtime/engine/ExternalTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixFailedJobCommandFactory.java
M backend/app/src/main/java/com/orqix/runtime/internal/ExternalJobService.java
M backend/app/src/main/java/com/orqix/runtime/internal/FailedJobRecorder.java
M backend/app/src/main/java/com/orqix/runtime/internal/JobFailure.java
A backend/app/src/main/java/com/orqix/runtime/internal/StepRetryPolicy.java
A backend/app/src/main/resources/db/migration/V220__connection_health_and_circuit.sql
M backend/app/src/test/java/com/orqix/resilience/Nodes.java
A backend/app/src/test/java/com/orqix/runtime/StepRetryAndCircuitTest.java
A backend/app/src/test/java/com/orqix/runtime/internal/StepRetryPolicyTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
```

```
M backend/app/src/test/java/com/orqix/support/RuntimeTestBase.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnDefaults.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/model/StepRetry.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/RetryRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/test/java/com/orqix/compiler/BpmnSourceTest.java
M backend/compiler/src/test/java/com/orqix/compiler/CompilerCasesTest.java
M backend/compiler/src/test/java/com/orqix/compiler/GoldenFilesTest.java
A backend/compiler/src/test/java/com/orqix/compiler/StepRetryTest.java
M backend/compiler/src/test/resources/golden/new-hire-onboarding/hashes.json
M backend/compiler/src/test/resources/golden/new-hire-onboarding/process.bpmn20.xml
M backend/compiler/src/test/resources/golden/order-payment-wait/hashes.json
M backend/compiler/src/test/resources/golden/order-payment-wait/process.bpmn20.xml
M backend/compiler/src/test/resources/golden/purchase-approval/hashes.json
M backend/compiler/src/test/resources/golden/purchase-approval/process.bpmn20.xml
M backend/compiler/src/test/resources/golden/start-date-onboarding/hashes.json
M backend/compiler/src/test/resources/golden/start-date-onboarding/process.bpmn20.xml
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/admin-guide.md
M docs/build/README.md
A docs/build/S131-retry-and-circuit-breaker.md
M docs/build/S4-workflow-compiler.md
M docs/user-guide.md
M examples/fixtures/compiler-cases.json
M frontend/src/api/schema.d.ts
A frontend/src/features/settings/ConnectionHealthPanel.test.tsx
A frontend/src/features/settings/ConnectionHealthPanel.tsx
M frontend/src/features/settings/ConnectionsPanel.tsx
A frontend/src/features/settings/connectionHealthApi.ts
M frontend/src/lib/events.test.ts
M frontend/src/lib/events.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/connectionHealth.ts
M frontend/src/test/server.ts
```

003ea821 feat: secret providers with access audit and worker business errors (S132)

```
M backend/app/src/main/java/com/orqix/integration/adapters/InboundService.java
M backend/app/src/main/java/com/orqix/integration/api/ConnectionDtos.java
A backend/app/src/main/java/com/orqix/integration/api/SecretAccessController.java
M backend/app/src/main/java/com/orqix/integration/internal/ConnectionAdminService.java
M backend/app/src/main/java/com/orqix/integration/internal/SecretStore.java
M backend/app/src/main/java/com/orqix/integration/kafka/KafkaConsumerRunner.java
A backend/app/src/main/java/com/orqix/integration/secrets/AwsSecretsManagerProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/AwsSigV4.java
A backend/app/src/main/java/com/orqix/integration/secrets/AzureKeyVaultProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/BearerTokens.java
A backend/app/src/main/java/com/orqix/integration/secrets/FileSecretProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/GoogleSecretManagerProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretAccessAudit.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretAccessQueries.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretHttp.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretProvider.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretProviderSettings.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretRef.java
A backend/app/src/main/java/com/orqix/integration/secrets/SecretResolver.java
A backend/app/src/main/java/com/orqix/integration/secrets/VaultSecretProvider.java
M backend/app/src/main/java/com/orqix/integration/webhooks/WebhookDispatcher.java
M backend/app/src/main/java/com/orqix/runtime/api/ExternalJobDtos.java
M backend/app/src/main/java/com/orqix/runtime/engine/ExternalTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/internal/ExternalJobService.java
M backend/app/src/main/resources/application.yaml
A backend/app/src/main/resources/db/migration/V240__secret_providers_and_access_audit.sql
A backend/app/src/main/resources/db/migration/V241__external_job_business_errors.sql
A backend/app/src/test/java/com/orqix/integration/secrets/SecretProvidersTest.java
A backend/app/src/test/java/com/orqix/integration/secrets/SecretProvidersVaultTest.java
A backend/app/src/test/java/com/orqix/runtime/ExternalWorkerErrorsTest.java
```

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FailureRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ScopeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M backend/compiler/src/test/java/com/orqix/compiler/GeneratedXmlTest.java
A backend/compiler/src/test/java/com/orqix/compiler/WorkerBusinessErrorsTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/admin-guide.md
A docs/build/S132-secret-providers-and-worker-errors.md
M frontend/src/api/schema.d.ts
M frontend/src/features/studio/StepNode.tsx
M frontend/src/features/studio/StepsView.tsx
M frontend/src/features/studio/StudioDialogs.tsx
M frontend/src/features/studio/inspector/ExternalTaskInspector.test.tsx
M frontend/src/features/studio/inspector/ExternalTaskInspector.tsx
M frontend/src/features/studio/inspector/ScopeInspectors.tsx
M frontend/src/features/studio/model/kinds.ts
M frontend/src/features/studio/model/ops.ts
M frontend/src/features/studio/notation/BpmnStepNode.tsx
M tools/worker-sdk/README.md
M tools/worker-sdk/orqix_worker.py
M tools/worker-sdk/test_orqix_worker.py
```

9f80e10b fix(integration): pass retryAfter in the paginated call result after the Wave A merge

```
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
```

49541af6 feat(waits): close S129 follow-ups after the Wave A merge

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/insights/api/InsightsController.java
M backend/app/src/main/java/com/orqix/insights/api/InsightsDtos.java
M backend/app/src/main/java/com/orqix/insights/internal/InsightsQueries.java
M backend/app/src/main/java/com/orqix/insights/internal/InsightsService.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
A backend/app/src/main/java/com/orqix/integration/ResultMappings.java
M backend/app/src/main/java/com/orqix/integration/V1Results.java
M backend/app/src/main/java/com/orqix/runtime/internal/OperatorService.java
M backend/app/src/main/java/com/orqix/runtime/internal/PollScheduler.java
M backend/app/src/main/java/com/orqix/runtime/internal/PollWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/ReleaseCompatibility.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunRepair.java
M backend/app/src/main/java/com/orqix/runtime/internal/WaitStates.java
A backend/app/src/main/resources/db/migration/V212__poll_wait_purge.sql
A backend/app/src/test/java/com/orqix/runtime/PollLoadTest.java
M backend/app/src/test/java/com/orqix/runtime/PollWaitsTest.java
M backend/app/src/test/java/com/orqix/support/TrackingStub.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/PollRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M backend/compiler/src/test/java/com/orqix/compiler/PollRulesTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/build/S129-check-until-and-waits.md
M frontend/src/api/schema.d.ts
M frontend/src/features/insights/InsightsPage.tsx
A frontend/src/features/insights/OutsideWaits.test.tsx
A frontend/src/features/insights/OutsideWaits.tsx
A frontend/src/features/studio/inspector/FallbackCheckFields.test.tsx
A frontend/src/features/studio/inspector/FallbackCheckFields.tsx
M frontend/src/features/studio/inspector/MessageInspector.tsx
M frontend/src/features/studio/inspector/PollFields.tsx
M frontend/src/features/studio/model/ops.ts
```

```
M frontend/src/features/studio/model/pollSchedule.ts
M frontend/src/mocks/simulator.ts
M frontend/src/mocks/waits.ts
```

1fc2d70b fix(db): renumber the poll purge migration to V242 and make it safe to rerun

```
R054 backend/app/src/main/resources/db/migration/V212__poll_wait_purge.sql
backend/app/src/main/resources/db/migration/V242__poll_wait_purge.sql
M docs/build/S129-check-until-and-waits.md
```

0ad8a224 feat(mcp): Orqix as an MCP server for AI assistants (CG-M04, S136)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/catalog/ToolWorkflows.java
M backend/app/src/main/java/com/orqix/identity/CallerResolver.java
A backend/app/src/main/java/com/orqix/mcp/McpAuthFilter.java
A backend/app/src/main/java/com/orqix/mcp/McpConfirmations.java
A backend/app/src/main/java/com/orqix/mcp/McpExplanations.java
A backend/app/src/main/java/com/orqix/mcp/McpProtocol.java
A backend/app/src/main/java/com/orqix/mcp/McpRateLimiter.java
A backend/app/src/main/java/com/orqix/mcp/McpResource.java
A backend/app/src/main/java/com/orqix/mcp/McpSchemas.java
A backend/app/src/main/java/com/orqix/mcp/McpSecurityConfiguration.java
A backend/app/src/main/java/com/orqix/mcp/McpServerSwitch.java
A backend/app/src/main/java/com/orqix/mcp/McpSessions.java
A backend/app/src/main/java/com/orqix/mcp/McpSettings.java
A backend/app/src/main/java/com/orqix/mcp/McpStreams.java
A backend/app/src/main/java/com/orqix/mcp/McpTokens.java
A backend/app/src/main/java/com/orqix/mcp/McpTools.java
A backend/app/src/main/java/com/orqix/mcp/api/McpEndpoint.java
A backend/app/src/main/java/com/orqix/mcp/api/McpMetadataController.java
A backend/app/src/main/java/com/orqix/mcp/api/McpServerController.java
A backend/app/src/main/java/com/orqix/mcp/package-info.java
A backend/app/src/main/java/com/orqix/runtime/AssistantRuntime.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunService.java
M backend/app/src/main/resources/application-test.yaml
M backend/app/src/main/resources/application.yaml
A backend/app/src/main/resources/db/migration/V280__mcp_server.sql
A backend/app/src/test/java/com/orqix/mcp/McpConfirmationTest.java
A backend/app/src/test/java/com/orqix/mcp/McpEndToEndTest.java
A backend/app/src/test/java/com/orqix/mcp/McpPermissionParityTest.java
A backend/app/src/test/java/com/orqix/mcp/McpProtocolTest.java
A backend/app/src/test/java/com/orqix/mcp/McpTestBase.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
A backend/app/src/test/java/com/orqix/support/McpTestClient.java
A backend/app/src/test/java/com/orqix/support/McpTestTokens.java
M backend/app/src/test/java/com/orqix/support/TestJwtDecoder.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M deploy/compose/keycloak/orqix-dev-realm.json
M docs/admin-guide.md
A docs/build/S136-orqix-mcp-server.md
M frontend/src/api/endpoints.ts
M frontend/src/api/schema.d.ts
M frontend/src/api/types.ts
A frontend/src/features/settings/McpServerPanel.test.tsx
A frontend/src/features/settings/McpServerPanel.tsx
M frontend/src/features/settings/SettingsPage.tsx
M frontend/src/features/studio/inspector/SimpleInspectors.tsx
M frontend/src/features/studio/model/types.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/mcpServer.ts
M frontend/src/test/server.ts
M frontend/src/test/setup.ts
```

515c5c4a feat(packages): connector packages, package triggers, async follow-up and the ServiceNow ITSM package (S133)

```
M .gitattributes
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/catalog/api/CatalogController.java
M backend/app/src/main/java/com/orqix/catalog/internal/WorkflowService.java
M backend/app/src/main/java/com/orqix/identity/AccessService.java
M backend/app/src/main/java/com/orqix/identity/Permission.java
M backend/app/src/main/java/com/orqix/identity/api/AccessController.java
M backend/app/src/main/java/com/orqix/identity/security/SecurityConfiguration.java
```

```
A backend/app/src/main/java/com/orqix/integration/AsyncStatus.java
A backend/app/src/main/java/com/orqix/integration/ConnectorPackageTemplates.java
M backend/app/src/main/java/com/orqix/integration/HttpExecutor.java
M backend/app/src/main/java/com/orqix/integration/V1Requests.java
M backend/app/src/main/java/com/orqix/integration/V1Results.java
A backend/app/src/main/java/com/orqix/integration/api/ConnectorPackageController.java
M backend/app/src/main/java/com/orqix/integration/internal/ConnectionAdminService.java
A backend/app/src/main/java/com/orqix/integration/packages/ConnectorPackageService.java
A backend/app/src/main/java/com/orqix/integration/packages/PackageArchive.java
A backend/app/src/main/java/com/orqix/integration/packages/PackagePolls.java
A backend/app/src/main/java/com/orqix/integration/packages/PackageSettings.java
A backend/app/src/main/java/com/orqix/integration/packages/PackageTriggerService.java
A backend/app/src/main/java/com/orqix/integration/packages/PackageValidator.java
A backend/app/src/main/resources/db/migration/V250__connector_packages.sql
A backend/app/src/test/java/com/orqix/integration/ConnectorPackageArchiveTest.java
A backend/app/src/test/java/com/orqix/integration/ConnectorPackageLifecycleTest.java
A backend/app/src/test/java/com/orqix/integration/PackageTriggersTest.java
A backend/app/src/test/java/com/orqix/runtime/AsyncStartFollowUpTest.java
A backend/app/src/test/java/com/orqix/runtime/ServiceNowPackageE2ETest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
A backend/app/src/test/java/com/orqix/support/ExamplePublisher.java
M backend/app/src/test/java/com/orqix/support/IntegrationTestBase.java
A backend/app/src/test/java/com/orqix/support/PackageTestBase.java
A backend/app/src/test/java/com/orqix/support/ReportJobStub.java
M backend/app/src/test/java/com/orqix/support/ServiceNowStub.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M backend/compiler/pom.xml
M backend/compiler/src/main/java/com/orqix/compiler/ConnectionContracts.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Connections.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M contracts/connection-definition-v1.schema.json
A contracts/connector-package-v1.schema.json
M contracts/openapi.yaml
A docs/build/S133-connector-packages.md
M examples/fixtures/connection-v1-cases.json
A examples/packages/servicenow-itsm-1.0.0.orqpkg
A examples/packages/servicenow-itsm/connections/servicenow-catalog.connection.json
A examples/packages/servicenow-itsm/connections/servicenow-itsm.connection.json
A examples/packages/servicenow-itsm/docs/README.md
A examples/packages/servicenow-itsm/fixtures/create-incident/created.json
A examples/packages/servicenow-itsm/fixtures/create-incident/rate-limited.json
A examples/packages/servicenow-itsm/fixtures/get-incident/in-progress.json
A examples/packages/servicenow-itsm/fixtures/get-incident/not-found.json
A examples/packages/servicenow-itsm/fixtures/get-incident/resolved.json
A examples/packages/servicenow-itsm/manifest.json
A examples/packages/servicenow-itsm/templates/steps/raise-incident.json
A examples/packages/servicenow-itsm/templates/steps/wait-until-resolved.json
A examples/packages/servicenow-itsm/templates/workflows/it-access-request.json
A examples/packages/servicenow-itsm/triggers/incident-changes.poll.json
A examples/packages/servicenow-itsm/triggers/incident-updated.webhook.json
M frontend/src/api/schema.d.ts
M frontend/src/features/settings/AccessPanel.tsx
A frontend/src/features/settings/ConnectorPackagesPanel.test.tsx
A frontend/src/features/settings/ConnectorPackagesPanel.tsx
M frontend/src/features/settings/SettingsPage.tsx
A frontend/src/features/settings/connectorPackagesApi.ts
M frontend/src/features/studio/Palette.tsx
M frontend/src/features/studio/StudioPage.tsx
A frontend/src/features/studio/inspector/HttpInspector.async.test.tsx
M frontend/src/features/studio/inspector/HttpInspector.tsx
A frontend/src/features/studio/model/packageSteps.test.ts
A frontend/src/features/studio/model/packageSteps.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/connectorPackages.ts
M frontend/src/mocks/simulator.ts
M frontend/src/test/server.ts
M tools/contract-check/consistency_gates.py
M tools/contract-check/contract_check.py
```

934618e8 test(studio): type the async variant in the call-and-wait inspector test

```
M frontend/src/features/studio/inspector/HttpInspector.async.test.tsx
```

e91666ea feat(ai): model provider SPI and the aiTask step in the compiler (CG-A04, CG-A01)

```
A backend/app/src/main/java/com/orqix/ai/provider/AnthropicProvider.java
A backend/app/src/main/java/com/orqix/ai/provider/AzureOpenAiProvider.java
A backend/app/src/main/java/com/orqix/ai/provider/BedrockProvider.java
A backend/app/src/main/java/com/orqix/ai/provider/ModelProvider.java
A backend/app/src/main/java/com/orqix/ai/provider/OpenAiCompatibleProvider.java
A backend/app/src/main/java/com/orqix/ai/provider/ProviderHttp.java
A backend/app/src/main/java/com/orqix/ai/provider/ProviderProperties.java
A backend/app/src/main/java/com/orqix/ai/provider/ProviderRegistry.java
A backend/app/src/main/java/com/orqix/ai/provider/SigV4.java
A backend/app/src/main/java/com/orqix/ai/provider/VertexGeminiProvider.java
A backend/app/src/main/java/com/orqix/integration/ModelProviderSecrets.java
M backend/compiler/build.gradle
M backend/compiler/pom.xml
M backend/compiler/src/main/java/com/orqix/compiler/CompileContext.java
M backend/compiler/src/main/java/com/orqix/compiler/CompiledWorkflow.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Enums.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/AiTaskRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/schema/SchemaValidation.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/AiTaskCompilerTest.java
M backend/compiler/src/test/java/com/orqix/compiler/GeneratedXmlTest.java
A contracts/ai-skill-v1.schema.json
M contracts/workflow-v0.schema.json
```

04fa2cd7 feat(ai): AI skills, guardrail pipeline, async AI jobs, evaluations, shadow mode and decision records (CG-A01, A02, A03, A05)

```
A backend/app/src/main/java/com/orqix/ai/api/AiSkillsController.java
A backend/app/src/main/java/com/orqix/ai/skills/AiSkills.java
A backend/app/src/main/java/com/orqix/ai/tasks/AiEvaluations.java
A backend/app/src/main/java/com/orqix/ai/tasks/AiShadow.java
A backend/app/src/main/java/com/orqix/ai/tasks/AiSkillsText.java
A backend/app/src/main/java/com/orqix/ai/tasks/AiTaskEngine.java
A backend/app/src/main/java/com/orqix/ai/tasks/AiTaskProperties.java
A backend/app/src/main/java/com/orqix/ai/tasks/Minimizer.java
A backend/app/src/main/java/com/orqix/ai/tasks/SkillAnswers.java
A backend/app/src/main/java/com/orqix/ai/tasks/TurnVault.java
M backend/app/src/main/java/com/orqix/catalog/internal/CompileContextFactory.java
M backend/app/src/main/java/com/orqix/catalog/internal/ReleaseService.java
M backend/app/src/main/java/com/orqix/identity/AccessService.java
M backend/app/src/main/java/com/orqix/identity/Permission.java
M backend/app/src/main/java/com/orqix/identity/api/AccessController.java
A backend/app/src/main/java/com/orqix/integration/AiTaskPort.java
A backend/app/src/main/java/com/orqix/runtime/api/AiDecisionsController.java
A backend/app/src/main/java/com/orqix/runtime/engine/AiTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixRoute.java
A backend/app/src/main/java/com/orqix/runtime/internal/AiDecisions.java
A backend/app/src/main/java/com/orqix/runtime/internal/AiTaskJobs.java
A backend/app/src/main/java/com/orqix/runtime/internal/AiTaskQueueSettings.java
A backend/app/src/main/java/com/orqix/runtime/internal/AiTaskWorker.java
M backend/app/src/main/java/com/orqix/shared/problem/ProblemCode.java
A backend/app/src/main/resources/db/migration/V260__ai_skills_and_decisions.sql
A backend/compiler/src/main/java/com/orqix/compiler/AiSkillContracts.java
M contracts/openapi.yaml
M frontend/src/lib/problemMessages.ts
```

01ef7e1b feat(ai): AiSkills API, evidence packs, AI task tests, Studio model for aiTask

```
M backend/app/src/main/java/com/orqix/ai/skills/AiSkills.java
M backend/app/src/main/java/com/orqix/ai/tasks/AiEvaluations.java
M backend/app/src/main/java/com/orqix/ai/tasks/AiTaskEngine.java
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/integration/AiTaskPort.java
```

```
M backend/app/src/main/java/com/orqix/runtime/engine/AiTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/evidencepack/ReleaseEvidencePack.java
M backend/app/src/main/java/com/orqix/runtime/evidencepack/RunEvidencePack.java
A backend/app/src/test/java/com/orqix/ai/AiTaskFlowTest.java
A backend/app/src/test/java/com/orqix/ai/provider/ModelProvidersTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
A backend/app/src/test/java/com/orqix/support/AiTaskFixtures.java
A backend/app/src/test/java/com/orqix/support/AiTaskStub.java
M backend/app/src/test/java/com/orqix/support/IntegrationTestBase.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/test/java/com/orqix/compiler/AiTaskCompilerTest.java
M contracts/openapi.yaml
M frontend/src/api/schema.d.ts
M frontend/src/features/settings/AccessPanel.tsx
M frontend/src/features/studio/RouteEdge.tsx
M frontend/src/features/studio/model/fieldRefs.ts
M frontend/src/features/studio/model/kinds.ts
M frontend/src/features/studio/model/notation.test.ts
M frontend/src/features/studio/model/notation.ts
M frontend/src/features/studio/model/summary.ts
M frontend/src/features/studio/model/types.ts
M frontend/src/features/studio/nodeMeta.ts
M frontend/src/mocks/handlers.ts
M frontend/src/mocks/simulator.ts
```

f8fea724 feat(studio): AI task inspector, AI decisions in run detail, mock handlers, AiSkills contract fixes

```
M contracts/openapi.yaml
M frontend/src/api/schema.d.ts
A frontend/src/features/runs/AiDecisionsPanel.test.tsx
A frontend/src/features/runs/AiDecisionsPanel.tsx
M frontend/src/features/runs/RunDetailPage.tsx
A frontend/src/features/studio/inspector/AiTaskInspector.test.tsx
A frontend/src/features/studio/inspector/AiTaskInspector.tsx
M frontend/src/features/studio/inspector/Inspector.tsx
M frontend/src/features/studio/model/notation.test.ts
A frontend/src/mocks/aiSkills.ts
M frontend/src/mocks/callbacks.ts
```

ad5067b0 docs(build): S134 AI decisions build note

```
A docs/build/S134-ai-decisions.md
```

47290b66 feat(agents): tool servers, durable agent loops, trace and agent tests (S135, CG-M01/M02/M03/M05/M07)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/integration/AgentExecutor.java
A backend/app/src/main/java/com/orqix/integration/AgentWaitTools.java
M backend/app/src/main/java/com/orqix/integration/ConnectionDefinition.java
M backend/app/src/main/java/com/orqix/integration/ConnectionService.java
M backend/app/src/main/java/com/orqix/integration/EffectDispatcher.java
M backend/app/src/main/java/com/orqix/integration/McpClient.java
A backend/app/src/main/java/com/orqix/integration/McpToolDriftDetected.java
M backend/app/src/main/java/com/orqix/integration/OAuthVault.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentDashboard.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentDecisionTools.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentKillSwitches.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentTests.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentToolRules.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentToolkit.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentTrace.java
A backend/app/src/main/java/com/orqix/integration/agenttools/AgentTraceRetention.java
A backend/app/src/main/java/com/orqix/integration/agenttools/McpServers.java
A backend/app/src/main/java/com/orqix/integration/agenttools/McpToolPins.java
A backend/app/src/main/java/com/orqix/integration/api/AgentToolsController.java
M backend/app/src/main/java/com/orqix/integration/internal/ConnectionAdminService.java
M backend/app/src/main/java/com/orqix/notification/NotificationService.java
A backend/app/src/main/java/com/orqix/runtime/api/AgentRequestsController.java
M backend/app/src/main/java/com/orqix/runtime/engine/EngineCommands.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
A backend/app/src/main/java/com/orqix/runtime/internal/AgentHumanRequests.java
A backend/app/src/main/java/com/orqix/runtime/internal/AgentLoopScheduler.java
A backend/app/src/main/java/com/orqix/runtime/internal/AgentLoops.java
A backend/app/src/main/java/com/orqix/runtime/internal/AgentToolNotifications.java
A backend/app/src/main/java/com/orqix/runtime/internal/AgentWaitToolsService.java
```

```
M backend/app/src/main/java/com/orqix/runtime/internal/ChildRuns.java
M backend/app/src/main/java/com/orqix/runtime/internal/FailedJobRecorder.java
A backend/app/src/main/resources/db/migration/V270__mcp_tool_snapshots.sql
A backend/app/src/main/resources/db/migration/V271__agent_kill_switches.sql
A backend/app/src/main/resources/db/migration/V272__agent_loops.sql
A backend/app/src/main/resources/db/migration/V273__agent_trace.sql
A backend/app/src/main/resources/db/migration/V274__agent_test_suites.sql
A backend/app/src/main/resources/db/migration/V275__agent_tool_notifications.sql
M backend/app/src/test/java/com/orqix/integration/McpClientTest.java
A backend/app/src/test/java/com/orqix/runtime/AgentLoopTest.java
A backend/app/src/test/java/com/orqix/runtime/AgentToolServersTest.java
A backend/app/src/test/java/com/orqix/runtime/AgentTraceAndTestsTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
A backend/app/src/test/java/com/orqix/support/AgentToolsTestBase.java
M backend/app/src/test/java/com/orqix/support/McpStub.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M contracts/connection-definition-v0.schema.json
M contracts/openapi.yaml
A docs/build/S135-agent-tools.md
A examples/connections/vendor-due-diligence.connection.json
A examples/connections/vendor-registry.mcp-server.json
M frontend/src/api/endpoints.ts
M frontend/src/api/schema.d.ts
M frontend/src/api/types.ts
M frontend/src/app/Shell.test.tsx
M frontend/src/app/Shell.tsx
M frontend/src/app/routes.tsx
A frontend/src/features/agenttools/AgentToolsPage.test.tsx
A frontend/src/features/agenttools/AgentToolsPage.tsx
M frontend/src/features/notifications/NotificationsMenu.tsx
M frontend/src/features/runs/RunDetailPage.tsx
A frontend/src/mocks/agentTools.ts
M frontend/src/mocks/browser.ts
M frontend/src/test/server.ts
```

6b611f5b fix: merged grantable permissions (V290) and mcpServer in the connection summary contract

```
A backend/app/src/main/resources/db/migration/V290__merge_grantable_permissions.sql
M backend/app/src/test/java/com/orqix/api/AccessApiTest.java
M contracts/openapi.yaml
M frontend/src/api/schema.d.ts
```

a55eaa31 docs(release): one change record PDF for Waves A and B

```
A docs/release/orqix-change-record-2026-11-06.pdf
D docs/release/wave-a-change-record-2026-11-06.pdf
```

e680c28a fix(contracts): known permission qualifiers and quoted inline descriptions in openapi.yaml

```
M contracts/openapi.yaml
M frontend/src/api/schema.d.ts
```

b419c3e9 feat(expr): Orqix expression language Java engine and shared cases (CG-X01)

```
M backend/compiler/src/main/java/com/orqix/compiler/BusinessCalendar.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Ast.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Checker.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Evaluator.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/ExprType.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Expression.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/ExpressionException.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Parser.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Scope.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Values.java
A backend/compiler/src/test/java/com/orqix/compiler/expr/ExpressionCasesTest.java
A examples/fixtures/expression-cases.json
```

7336909c fix(packages): time-zone independent connector package archives

```
M backend/app/src/main/java/com/orqix/integration/packages/PackageArchive.java
M examples/packages/servicenow-itsm-1.0.0.orqpkg
```

1904e3dc feat(compiler): S139 working-time waits, reminders, wait for the first, message handlers

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnDefaults.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
```

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Enums.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/model/WaitExtras.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/model/WaitsReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/PollRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ScopeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TimerRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/WaitsAndStartsRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M backend/compiler/src/test/java/com/orqix/compiler/BpmnSourceScopesTest.java
A backend/compiler/src/test/java/com/orqix/compiler/WaitsAndStartsCompilerTest.java
M contracts/workflow-v0.schema.json
A examples/order-payment-first.json
```

3662d059 feat(expr): Calculate step, expression conditions, = mappings, computed form fields and rules in the compiler (CG-X01, CG-F04)

```
A backend/app/src/main/java/com/orqix/runtime/engine/CalculateTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/CompensationTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixRoute.java
M backend/app/src/main/java/com/orqix/runtime/engine/PollTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/ProcessData.java
M backend/app/src/main/java/com/orqix/runtime/internal/IncidentCorrectionService.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunPathBuilder.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunService.java
A backend/compiler/src/main/java/com/orqix/compiler/FormComputation.java
M backend/compiler/src/main/java/com/orqix/compiler/FormVisibility.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/Calculation.java
A backend/compiler/src/main/java/com/orqix/compiler/expr/PinnedTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/eval/DefaultFormValidator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Enums.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ExpressionRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FormGuards.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ScopeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/StageRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/schema/SchemaValidation.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/CalculateCompilerTest.java
M backend/compiler/src/test/java/com/orqix/compiler/GeneratedXmlTest.java
A backend/compiler/src/test/java/com/orqix/compiler/expr/ExpressionApiTest.java
M contracts/workflow-v0.schema.json
A frontend/src/lib/expr/checker.ts
A frontend/src/lib/expr/dates.ts
A frontend/src/lib/expr/decimal.ts
A frontend/src/lib/expr/evaluator.ts
A frontend/src/lib/expr/expr.test.ts
A frontend/src/lib/expr/index.ts
A frontend/src/lib/expr/parser.ts
A frontend/src/lib/expr/types.ts
```

ca847361 fix(mcp, tests): report truncated workflow lists; stop package tests leaking grants

```
M backend/app/src/main/java/com/orqix/mcp/McpTools.java
M backend/app/src/test/java/com/orqix/mcp/McpPermissionParityTest.java
```

```
M backend/app/src/test/java/com/orqix/support/PackageTestBase.java
```

d326af06 feat(expr): engine delegate for Calculate, expression routes and = mappings, server-computed form fields, evaluateExpression API (CG-X01, CG-F04)

```
A backend/app/src/main/java/com/orqix/catalog/api/ExpressionController.java
M backend/app/src/main/java/com/orqix/runtime/engine/CalculateTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixRoute.java
M backend/app/src/main/java/com/orqix/runtime/engine/PollTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/ProcessData.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunService.java
A backend/app/src/test/java/com/orqix/runtime/ExpressionsRuntimeTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
A backend/compiler/src/main/java/com/orqix/compiler/Expressions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/DefaultWorkflowSimulator.java
M contracts/openapi.yaml
M docs/build/S3-api.md
M frontend/src/api/endpoints.ts
M frontend/src/api/schema.d.ts
M tools/contract-check/contract_check.py
```

ae0af1b5 feat(runtime): S139 calendars, reminders, wait for the first, message handlers, scheduled starts

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/catalog/ReleaseQueries.java
A backend/app/src/main/java/com/orqix/catalog/WaitsAndStartsNodes.java
M backend/app/src/main/java/com/orqix/catalog/internal/AssetReferenceCheck.java
A backend/app/src/main/java/com/orqix/runtime/api/WaitsAndStartsController.java
M backend/app/src/main/java/com/orqix/runtime/engine/OrqixWait.java
M backend/app/src/main/java/com/orqix/runtime/engine/RuntimeProjectionListener.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageDelivery.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageService.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageStore.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/PollWaits.java
A backend/app/src/main/java/com/orqix/runtime/internal/Recurrence.java
A backend/app/src/main/java/com/orqix/runtime/internal/Schedules.java
A backend/app/src/main/java/com/orqix/runtime/internal/WaitReminders.java
M backend/app/src/main/java/com/orqix/runtime/internal/WaitStates.java
A backend/app/src/main/java/com/orqix/runtime/internal/WaitsAndStartsScheduler.java
A backend/app/src/main/java/com/orqix/runtime/internal/WorkingCalendars.java
M backend/app/src/main/resources/application-test.yaml
A backend/app/src/main/resources/db/migration/V320__working_calendars.sql
A backend/app/src/main/resources/db/migration/V321__wait_reminders_and_message_handlers.sql
A backend/app/src/main/resources/db/migration/V322__schedule_triggers.sql
A backend/app/src/test/java/com/orqix/runtime/WaitsAndStartsRuntimeTest.java
A backend/app/src/test/java/com/orqix/runtime/internal/RecurrenceTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M contracts/openapi.yaml
M tools/contract-check/contract_check.py
```

b015942b test(compiler): S139 BPMN round trips and remaining rule paths; optional correlation fields refused

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/WaitsAndStartsRules.java
A backend/compiler/src/test/java/com/orqix/compiler/WaitsAndStartsRoundTripTest.java
```

d6bb4df0 docs(build): S134 verification results

```
M docs/build/S134-ai-decisions.md
```

29b0a045 test(e2e): wait for each assistant reply; expect the v1 OpenAPI path-parameter reason

```
M tests/e2e/specs/t-in-01-openapi-import.spec.ts
M tests/e2e/specs/t-ui-12-conversation.spec.ts
```

9c65ef79 fix(agents): allow kind mcpServer in the connection list contract; record S135 evidence

```
M contracts/openapi.yaml
M docs/build/S135-agent-tools.md
M frontend/src/api/schema.d.ts
```

546ef2db feat(studio): Calculate inspector, guided expression editor, formula paths, computed form fields and rules in the browser and mocks; S137 note

```
A docs/build/S137-expressions-and-calculate.md
M frontend/src/components/form/FormRenderer.tsx
M frontend/src/features/studio/inspector/BranchInspector.tsx
A frontend/src/features/studio/inspector/CalculateInspector.test.tsx
A frontend/src/features/studio/inspector/CalculateInspector.tsx
A frontend/src/features/studio/inspector/ExpressionEditor.tsx
M frontend/src/features/studio/inspector/Inspector.tsx
A frontend/src/features/studio/model/conditionText.ts
A frontend/src/features/studio/model/expressionScope.ts
M frontend/src/features/studio/model/fieldRefs.ts
M frontend/src/features/studio/model/kinds.ts
M frontend/src/features/studio/model/notation.test.ts
M frontend/src/features/studio/model/notation.ts
M frontend/src/features/studio/model/summary.ts
M frontend/src/features/studio/model/types.ts
M frontend/src/features/studio/nodeMeta.ts
A frontend/src/lib/expr/calculate.ts
M frontend/src/lib/expr/checker.ts
A frontend/src/lib/expr/describe.test.ts
A frontend/src/lib/expr/describe.ts
A frontend/src/lib/form/computed.test.ts
A frontend/src/lib/form/computed.ts
M frontend/src/lib/form/schema.ts
M frontend/src/lib/form/validate.ts
M frontend/src/lib/form/values.ts
M frontend/src/lib/form/visibility.ts
M frontend/src/mocks/engine.ts
M frontend/src/mocks/handlers.ts
M frontend/src/mocks/simulator.ts
M frontend/src/mocks/validate.ts
```

3f8e2172 feat(studio,settings): S139 inspectors, calendars and schedules UI; build note

```
M contracts/openapi.yaml
A docs/build/S139-waits-reminders-schedules.md
M frontend/src/api/schema.d.ts
A frontend/src/features/settings/SchedulesPanel.tsx
M frontend/src/features/settings/SettingsPage.tsx
A frontend/src/features/settings/WorkingCalendarsPanel.tsx
M frontend/src/features/studio/RouteEdge.tsx
M frontend/src/features/studio/StepNode.tsx
M frontend/src/features/studio/inspector/Inspector.tsx
M frontend/src/features/studio/inspector/ScopeInspectors.tsx
M frontend/src/features/studio/inspector/SimpleInspectors.tsx
A frontend/src/features/studio/inspector/WaitsAndStartsFields.tsx
M frontend/src/features/studio/model/fieldRefs.ts
M frontend/src/features/studio/model/kinds.ts
M frontend/src/features/studio/model/notation.test.ts
M frontend/src/features/studio/model/notation.ts
M frontend/src/features/studio/model/ops.ts
M frontend/src/features/studio/model/summary.ts
M frontend/src/features/studio/model/types.ts
A frontend/src/features/studio/model/waitsAndStarts.ts
M frontend/src/features/studio/nodeMeta.ts
M frontend/src/features/studio/notation/BpmnGlyph.tsx
M frontend/src/mocks/callbacks.ts
M frontend/src/mocks/simulator.ts
A frontend/src/mocks/waitsAndStarts.ts
M frontend/src/test/setup.ts
```

39ce7897 feat(S141): escalation policies, actionable tasks in Teams/Slack/email, workspace step templates

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/catalog/api/StepTemplateController.java
A backend/app/src/main/java/com/orqix/catalog/internal/StepTemplates.java
M backend/app/src/main/java/com/orqix/identity/security/SecurityConfiguration.java
A backend/app/src/main/java/com/orqix/integration/ChannelSecrets.java
M backend/app/src/main/java/com/orqix/notification/NotificationService.java
A backend/app/src/main/java/com/orqix/runtime/api/TaskActionController.java
M backend/app/src/main/java/com/orqix/runtime/engine/RuntimeProjectionListener.java
A backend/app/src/main/java/com/orqix/runtime/internal/LocalTaskActionStub.java
A backend/app/src/main/java/com/orqix/runtime/internal/TaskActionLinks.java
A backend/app/src/main/java/com/orqix/runtime/internal/TaskActionOutbox.java
A backend/app/src/main/java/com/orqix/runtime/internal/TaskActionService.java
A backend/app/src/main/java/com/orqix/runtime/internal/TaskActionSettings.java
A backend/app/src/main/java/com/orqix/runtime/internal/TaskActionTransport.java
A backend/app/src/main/java/com/orqix/runtime/internal/TaskEscalations.java
```

```
M backend/app/src/main/java/com/orqix/runtime/internal/TaskService.java
M backend/app/src/main/java/com/orqix/runtime/internal/TaskStore.java
A backend/app/src/main/resources/db/migration/V340__task_escalation_policies.sql
A backend/app/src/main/resources/db/migration/V341__actionable_tasks.sql
A backend/app/src/main/resources/db/migration/V342__workspace_step_templates.sql
A backend/app/src/test/java/com/orqix/catalog/StepTemplatesApiTest.java
A backend/app/src/test/java/com/orqix/runtime/ActionableTasksTest.java
A backend/app/src/test/java/com/orqix/runtime/TaskEscalationTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/Pipeline.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/EscalationPolicyRules.java
A backend/compiler/src/test/java/com/orqix/compiler/EscalationPolicyRulesTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M docs/admin-guide.md
A docs/build/S141-escalation-actionable-tasks-templates.md
M docs/user-guide.md
M frontend/src/api/endpoints.ts
M frontend/src/api/schema.d.ts
M frontend/src/api/types.ts
M frontend/src/app/routes.tsx
M frontend/src/features/settings/SettingsPage.tsx
A frontend/src/features/settings/TaskActionsPanel.test.tsx
A frontend/src/features/settings/TaskActionsPanel.tsx
M frontend/src/features/studio/Palette.tsx
M frontend/src/features/studio/StudioPage.tsx
A frontend/src/features/studio/WorkspaceTemplates.tsx
A frontend/src/features/studio/inspector/EscalationPolicyFields.test.tsx
A frontend/src/features/studio/inspector/EscalationPolicyFields.tsx
M frontend/src/features/studio/inspector/Inspector.tsx
M frontend/src/features/studio/inspector/WorkManagementFields.tsx
A frontend/src/features/studio/model/workspaceTemplates.test.ts
A frontend/src/features/studio/model/workspaceTemplates.ts
A frontend/src/features/tasks/TaskActionLinkPage.test.tsx
A frontend/src/features/tasks/TaskActionLinkPage.tsx
M frontend/src/lib/events.test.ts
M frontend/src/lib/events.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/taskActions.ts
M frontend/src/test/server.ts
M frontend/src/test/setup.ts
M tools/contract-check/consistency_gates.py
```

52c2667b feat(forms): CG-F01 time/date-time/checklist/tags, CG-F02 lookup fields, CG-F05 guest response forms (S140)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/identity/security/SecurityConfiguration.java
A backend/app/src/main/java/com/orqix/integration/FormLookupCalls.java
A backend/app/src/main/java/com/orqix/runtime/api/GuestDtos.java
A backend/app/src/main/java/com/orqix/runtime/api/GuestResponseController.java
A backend/app/src/main/java/com/orqix/runtime/api/LookupDtos.java
A backend/app/src/main/java/com/orqix/runtime/internal/FormLookups.java
A backend/app/src/main/java/com/orqix/runtime/internal/GuestLinkSender.java
A backend/app/src/main/java/com/orqix/runtime/internal/GuestResponseService.java
A backend/app/src/main/java/com/orqix/runtime/internal/GuestResponseStore.java
M backend/app/src/main/java/com/orqix/runtime/internal/MessageWaits.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunAttachments.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunService.java
M backend/app/src/main/java/com/orqix/shared/problem/ProblemCode.java
M backend/app/src/main/java/com/orqix/shared/web/SpaController.java
A backend/app/src/main/resources/db/migration/V330__guest_response_forms.sql
A backend/app/src/test/java/com/orqix/runtime/GuestResponseFormsTest.java
A backend/app/src/test/java/com/orqix/runtime/LookupFieldsTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/dmn/DmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/eval/DefaultFormValidator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Enums.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
```

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Node.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Conditions.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FormFieldRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/StepForms.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/LookupGuestFieldTypesTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
A docs/build/S140-lookup-guest-forms-field-types.md
A examples/connections/servicenow-users.connection.json
A frontend/src/api/guest.ts
M frontend/src/api/schema.d.ts
M frontend/src/app/routes.tsx
A frontend/src/components/form/FormRenderer.s140.test.tsx
M frontend/src/components/form/FormRenderer.tsx
A frontend/src/components/form/LookupField.tsx
A frontend/src/features/guest/GuestResponsePage.test.tsx
A frontend/src/features/guest/GuestResponsePage.tsx
A frontend/src/features/runs/GuestLinksPanel.tsx
M frontend/src/features/runs/RunDetailPage.tsx
M frontend/src/features/start/StartRequestPage.tsx
M frontend/src/features/studio/inspector/FieldListEditor.tsx
A frontend/src/features/studio/inspector/GuestFormSection.tsx
M frontend/src/features/studio/inspector/MessageInspector.tsx
M frontend/src/features/studio/model/ops.ts
M frontend/src/features/studio/model/types.ts
M frontend/src/lib/events.ts
M frontend/src/lib/form/display.ts
A frontend/src/lib/form/s140.test.ts
M frontend/src/lib/form/schema.ts
M frontend/src/lib/form/validate.ts
M frontend/src/lib/form/values.ts
A frontend/src/lib/form/zoned.ts
M frontend/src/lib/problemMessages.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/guestForms.ts
M frontend/src/test/server.ts
M tools/contract-check/consistency_gates.py
M tools/contract-check/contract_check.py
```

2e1137cc feat(control-flow): bounded loops (CG-E01) and for each on any step (CG-E06) - S138

```
A backend/app/src/main/java/com/orqix/runtime/api/StepItemsController.java
M backend/app/src/main/java/com/orqix/runtime/engine/AdapterTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/AiTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/EngineCommands.java
M backend/app/src/main/java/com/orqix/runtime/engine/ExternalTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/FlowableEngineConfiguration.java
M backend/app/src/main/java/com/orqix/runtime/engine/HttpTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/engine/NotificationTaskDelegate.java
A backend/app/src/main/java/com/orqix/runtime/engine/OrqixForEach.java
A backend/app/src/main/java/com/orqix/runtime/engine/OrqixLoop.java
M backend/app/src/main/java/com/orqix/runtime/engine/ProcessData.java
M backend/app/src/main/java/com/orqix/runtime/engine/RuntimeProjectionListener.java
M backend/app/src/main/java/com/orqix/runtime/internal/ReleaseCompatibility.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunPathBuilder.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunPaths.java
M backend/app/src/main/java/com/orqix/runtime/internal/TaskStore.java
A backend/app/src/main/resources/db/migration/V310__loops_and_for_each.sql
A backend/app/src/test/java/com/orqix/runtime/LoopsAndForEachRuntimeTest.java
M backend/app/src/test/java/com/orqix/runtime/ReleaseCompatibilityTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/Enums.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/model/Loops.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/model/StepForEach.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/AiTaskRules.java
```

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ForEachRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Graph.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/GraphRules.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/LoopRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
M backend/compiler/src/test/java/com/orqix/compiler/GeneratedXmlTest.java
A backend/compiler/src/test/java/com/orqix/compiler/LoopsAndForEachTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
A docs/build/S138-loops-and-for-each.md
M docs/user-guide.md
M frontend/src/api/schema.d.ts
M frontend/src/features/runs/diagram/RunDiagram.tsx
M frontend/src/features/runs/diagram/api.ts
A frontend/src/features/runs/diagram/loopsAndItems.test.ts
M frontend/src/features/runs/diagram/pathModel.ts
M frontend/src/features/studio/RouteEdge.tsx
M frontend/src/features/studio/inspector/AiTaskInspector.tsx
M frontend/src/features/studio/inspector/ExternalTaskInspector.tsx
A frontend/src/features/studio/inspector/ForEachFields.tsx
M frontend/src/features/studio/inspector/HttpInspector.tsx
M frontend/src/features/studio/inspector/Inspector.tsx
A frontend/src/features/studio/inspector/SendBackFields.test.tsx
A frontend/src/features/studio/inspector/SendBackFields.tsx
M frontend/src/features/studio/model/graph.ts
M frontend/src/features/studio/model/kinds.ts
M frontend/src/features/studio/model/ops.ts
M frontend/src/features/studio/model/summary.ts
M frontend/src/features/studio/model/types.ts
M frontend/src/mocks/simulator.ts
```

245d172c fix(expr): keep time and date-time fields out of formulas after the Wave C merge

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ExpressionRules.java
```

0f3f995d test(studio): 22 step kinds after the Wave C merge (calculate and waitForFirst)

```
M frontend/src/features/studio/model/notation.test.ts
```

4aa6eadb fix(contracts): quote two schedule and calendar descriptions with commas (Redocly)

```
M contracts/openapi.yaml
M frontend/src/api/schema.d.ts
```

5cd4185b feat(ai): document-type templates and the document member of the AI skill contract (CG-A06)

```
M backend/compiler/src/main/java/com/orqix/compiler/AiSkillContracts.java
A backend/compiler/src/main/java/com/orqix/compiler/DocumentTemplates.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/AiTaskRules.java
A backend/compiler/src/test/java/com/orqix/compiler/DocumentTemplatesTest.java
M contracts/ai-skill-v1.schema.json
```

8609adec feat(ai-skills): Studio pages for AI skills, evaluations, shadow, promotions, review queue and AI policy (S146 part D)

```
A docs/build/S146-parts/D-ai-skills-studio.md
M frontend/src/app/Shell.test.tsx
M frontend/src/app/Shell.tsx
M frontend/src/app/routes.tsx
A frontend/src/features/aiskills/AiPolicyAndReview.test.tsx
A frontend/src/features/aiskills/AiPolicyPage.tsx
A frontend/src/features/aiskills/AiProblem.tsx
A frontend/src/features/aiskills/AiReviewPage.tsx
A frontend/src/features/aiskills/AiSkillDetailPage.test.tsx
A frontend/src/features/aiskills/AiSkillDetailPage.tsx
A frontend/src/features/aiskills/AiSkillsPage.test.tsx
A frontend/src/features/aiskills/AiSkillsPage.tsx
A frontend/src/features/aiskills/EvalSuitesPanel.tsx
A frontend/src/features/aiskills/EvaluationPanel.tsx
A frontend/src/features/aiskills/PromotionsPanel.tsx
A frontend/src/features/aiskills/ShadowPanel.tsx
A frontend/src/features/aiskills/SkillEditor.tsx
A frontend/src/features/aiskills/api.ts
A frontend/src/features/aiskills/skillDocument.test.ts
```

```
A frontend/src/features/aiskills/skillDocument.ts
M frontend/src/mocks/aiSkills.ts
M frontend/src/mocks/state.ts
```

9ef01db3 fix(runtime): reminder domain check uses the shared JSON parser, not a compiler internal

```
M backend/app/src/main/java/com/orqix/runtime/internal/WaitReminders.java
```

12f6a064 feat(ai): OCR providers, document extraction pipeline and verification tasks (CG-A06)

```
A backend/app/src/main/java/com/orqix/ai/api/DocumentExtractionController.java
A backend/app/src/main/java/com/orqix/ai/ocr/AzureDocumentIntelligenceOcr.java
A backend/app/src/main/java/com/orqix/ai/ocr/LocalOcr.java
A backend/app/src/main/java/com/orqix/ai/ocr/OcrHttp.java
A backend/app/src/main/java/com/orqix/ai/ocr/OcrProperties.java
A backend/app/src/main/java/com/orqix/ai/ocr/OcrProviders.java
A backend/app/src/main/java/com/orqix/ai/ocr/TesseractOcr.java
A backend/app/src/main/java/com/orqix/ai/ocr/TextractOcr.java
M backend/app/src/main/java/com/orqix/ai/provider/SigV4.java
M backend/app/src/main/java/com/orqix/ai/tasks/AiEvaluations.java
M backend/app/src/main/java/com/orqix/ai/tasks/AiTaskEngine.java
A backend/app/src/main/java/com/orqix/ai/tasks/DocumentStage.java
M backend/app/src/main/java/com/orqix/ai/tasks/Minimizer.java
M backend/app/src/main/java/com/orqix/ai/tasks/SkillAnswers.java
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/integration/AiTaskPort.java
A backend/app/src/main/java/com/orqix/integration/OcrDocument.java
A backend/app/src/main/java/com/orqix/integration/OcrException.java
M backend/app/src/main/java/com/orqix/integration/OcrProvider.java
A backend/app/src/main/java/com/orqix/integration/RunDocuments.java
A backend/app/src/main/java/com/orqix/runtime/api/AiVerificationsController.java
M backend/app/src/main/java/com/orqix/runtime/engine/AiTaskDelegate.java
M backend/app/src/main/java/com/orqix/runtime/internal/AiTaskJobs.java
A backend/app/src/main/java/com/orqix/runtime/internal/AiVerifications.java
M backend/app/src/main/java/com/orqix/runtime/internal/AttachmentLifecycle.java
A backend/app/src/main/java/com/orqix/runtime/internal/RunDocumentsAdapter.java
A backend/app/src/main/resources/db/migration/V390__document_extraction.sql
A backend/app/src/test/java/com/orqix/ai/ocr/OcrProvidersTest.java
```

9be364a5 wip: S142 bulk operations, data search and job dashboard

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/identity/AccessService.java
M backend/app/src/main/java/com/orqix/identity/Permission.java
M backend/app/src/main/java/com/orqix/identity/api/AccessController.java
A backend/app/src/main/java/com/orqix/integration/IntegrationStatus.java
M backend/app/src/main/java/com/orqix/integration/health/ConnectionHealthService.java
A backend/app/src/main/java/com/orqix/integration/health/IntegrationStatusService.java
A backend/app/src/main/java/com/orqix/runtime/api/BulkAndSearchController.java
A backend/app/src/main/java/com/orqix/runtime/api/BulkDtos.java
M backend/app/src/main/java/com/orqix/runtime/api/RuntimeController.java
M backend/app/src/main/java/com/orqix/runtime/api/RuntimeDtos.java
A backend/app/src/main/java/com/orqix/runtime/internal/BulkOperationWorker.java
A backend/app/src/main/java/com/orqix/runtime/internal/BulkOperations.java
M backend/app/src/main/java/com/orqix/runtime/internal/ExternalJobService.java
M backend/app/src/main/java/com/orqix/runtime/internal/FailedJobRecorder.java
A backend/app/src/main/java/com/orqix/runtime/internal/JobDashboard.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunQueries.java
A backend/app/src/main/java/com/orqix/runtime/internal/RunSearchIndex.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunService.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunStore.java
M backend/app/src/main/java/com/orqix/runtime/internal/RuntimeViews.java
M backend/app/src/main/java/com/orqix/shared/problem/ProblemCode.java
A backend/app/src/main/resources/db/migration/V350__run_bulk_operate_permission.sql
A backend/app/src/main/resources/db/migration/V351__run_search_values.sql
A backend/app/src/main/resources/db/migration/V352__bulk_operations.sql
A backend/app/src/main/resources/db/migration/V353__external_worker_seen.sql
M backend/app/src/test/java/com/orqix/api/AccessApiTest.java
A backend/app/src/test/java/com/orqix/runtime/internal/BulkAndSearchTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/Pipeline.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/rules/SearchableFieldRules.java
A backend/compiler/src/test/java/com/orqix/compiler/SearchableFieldRulesTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
M frontend/src/api/endpoints.ts
```

```
M frontend/src/api/schema.d.ts
A frontend/src/features/runs/BulkActionsPanel.tsx
A frontend/src/features/runs/FieldFilters.tsx
M frontend/src/features/runs/RunsPage.tsx
A frontend/src/features/runs/bulkApi.ts
M frontend/src/features/settings/AccessPanel.tsx
A frontend/src/features/settings/JobDashboardPanel.tsx
M frontend/src/features/settings/SettingsPage.tsx
M frontend/src/lib/problemMessages.ts
M frontend/src/lib/useListFilters.ts
M frontend/src/mocks/browser.ts
A frontend/src/mocks/bulk.ts
M frontend/src/mocks/handlers.ts
M frontend/src/mocks/state.ts
M frontend/src/test/server.ts
M frontend/src/test/setup.ts
```

b5112470 feat(S144): alert rules, run retention and archiving, engine history purge (CG-O06, CG-O07, CG-P04)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
M backend/app/src/main/java/com/orqix/identity/AccessService.java
M backend/app/src/main/java/com/orqix/identity/Permission.java
M backend/app/src/main/java/com/orqix/identity/api/AccessController.java
A backend/app/src/main/java/com/orqix/integration/AlertChannels.java
A backend/app/src/main/java/com/orqix/integration/alerts/AlertChannelService.java
A backend/app/src/main/java/com/orqix/integration/alerts/AlertHookTransport.java
A backend/app/src/main/java/com/orqix/integration/alerts/LocalAlertHookStub.java
M backend/app/src/main/java/com/orqix/integration/webhooks/WebhookService.java
A backend/app/src/main/java/com/orqix/operations/alerts/AlertEngine.java
A backend/app/src/main/java/com/orqix/operations/alerts/AlertService.java
A backend/app/src/main/java/com/orqix/operations/alerts/AlertSignals.java
A backend/app/src/main/java/com/orqix/operations/api/AlertController.java
A backend/app/src/main/java/com/orqix/operations/api/RetentionController.java
A backend/app/src/main/java/com/orqix/operations/package-info.java
A backend/app/src/main/java/com/orqix/operations/retention/ArchiveStore.java
A backend/app/src/main/java/com/orqix/operations/retention/FileSystemArchiveStore.java
A backend/app/src/main/java/com/orqix/operations/retention/HistoryPurger.java
A backend/app/src/main/java/com/orqix/operations/retention/RetentionScheduling.java
A backend/app/src/main/java/com/orqix/operations/retention/RetentionService.java
A backend/app/src/main/java/com/orqix/operations/retention/RetentionSettings.java
A backend/app/src/main/java/com/orqix/operations/retention/RetentionSweeper.java
A backend/app/src/main/java/com/orqix/operations/retention/RunArchiver.java
A backend/app/src/main/java/com/orqix/operations/retention/S3ArchiveStore.java
A backend/app/src/main/java/com/orqix/runtime/RunRetention.java
A backend/app/src/main/java/com/orqix/runtime/engine/EngineHistoryPurge.java
M backend/app/src/main/java/com/orqix/runtime/evidencepack/RunEvidencePack.java
M backend/app/src/main/resources/application-test.yaml
M backend/app/src/main/resources/application.yaml
A backend/app/src/main/resources/db/migration/V370__alert_rules.sql
A backend/app/src/main/resources/db/migration/V371__run_retention.sql
M backend/app/src/test/java/com/orqix/api/AccessApiTest.java
A backend/app/src/test/java/com/orqix/operations/AlertRulesTest.java
A backend/app/src/test/java/com/orqix/operations/ArchiveStoresTest.java
A backend/app/src/test/java/com/orqix/operations/RunRetentionTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M contracts/openapi.yaml
M docs/admin-guide.md
A docs/build/S144-alerts-retention-archiving.md
M frontend/src/api/endpoints.ts
M frontend/src/api/schema.d.ts
M frontend/src/api/types.ts
M frontend/src/features/settings/AccessPanel.tsx
A frontend/src/features/settings/AlertRulesPanel.test.tsx
A frontend/src/features/settings/AlertRulesPanel.tsx
A frontend/src/features/settings/RunRetentionPanel.test.tsx
A frontend/src/features/settings/RunRetentionPanel.tsx
M frontend/src/features/settings/SettingsPage.tsx
A frontend/src/mocks/alertsRetention.ts
M frontend/src/mocks/browser.ts
M frontend/src/mocks/handlers.ts
M frontend/src/mocks/state.ts
M frontend/src/test/server.ts
M frontend/src/test/setup.ts
```

b6481547 feat(runtime): S146 part C - "Explain this request" for operators (CG-A07)

```
A backend/app/src/main/java/com/orqix/ai/tasks/OperatorExplainModel.java
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/integration/ConnectionStatus.java
A backend/app/src/main/java/com/orqix/integration/OperatorExplainPort.java
A backend/app/src/main/java/com/orqix/runtime/api/ExplainDtos.java
A backend/app/src/main/java/com/orqix/runtime/api/RunExplainController.java
A backend/app/src/main/java/com/orqix/runtime/explain/AiAnswer.java
A backend/app/src/main/java/com/orqix/runtime/explain/ExplainProperties.java
A backend/app/src/main/java/com/orqix/runtime/explain/RuleExplainer.java
A backend/app/src/main/java/com/orqix/runtime/explain/RunExplainService.java
A backend/app/src/main/java/com/orqix/runtime/explain/RunFacts.java
A backend/app/src/main/resources/db/migration/V394__ai_operator_explain.sql
A backend/app/src/test/java/com/orqix/runtime/RunExplainTest.java
A backend/app/src/test/java/com/orqix/runtime/explain/RunFactsTest.java
M backend/app/src/test/java/com/orqix/support/AiTaskFixtures.java
M backend/app/src/test/java/com/orqix/support/AiTaskStub.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M contracts/openapi.yaml
A docs/build/S146-parts/C-operator-explain.md
M frontend/src/api/schema.d.ts
A frontend/src/features/runs/ExplainRunPanel.test.tsx
A frontend/src/features/runs/ExplainRunPanel.tsx
M frontend/src/features/runs/RunDetailPage.tsx
M frontend/src/mocks/handlers.ts
A frontend/src/mocks/runExplain.ts
```

4c6ac65e fix: quote two OpenAPI descriptions and keep WaitReminders off compiler internals

```
M backend/app/src/main/java/com/orqix/runtime/internal/WaitReminders.java
M contracts/openapi.yaml
```

60c3a30a feat(ai): document extraction API, verification screen, tests and S146 part B notes (CG-A06)

```
A backend/app/src/test/java/com/orqix/ai/DocumentExtractionFlowTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M backend/app/src/test/java/com/orqix/support/AiTaskFixtures.java
M backend/app/src/test/java/com/orqix/support/AiTaskStub.java
A backend/app/src/test/java/com/orqix/support/OcrStub.java
M backend/app/src/test/java/com/orqix/support/Stub.java
M contracts/openapi.yaml
A docs/build/S146-parts/B-document-extraction.md
M frontend/src/api/schema.d.ts
M frontend/src/app/routes.tsx
A frontend/src/features/runs/DocumentVerificationsPanel.tsx
M frontend/src/features/runs/RunDetailPage.tsx
M frontend/src/features/studio/inspector/AiTaskInspector.test.tsx
M frontend/src/features/studio/inspector/AiTaskInspector.tsx
A frontend/src/features/verification/DocumentPreview.tsx
A frontend/src/features/verification/VerificationPage.test.tsx
A frontend/src/features/verification/VerificationPage.tsx
M frontend/src/mocks/callbacks.ts
A frontend/src/mocks/documentExtraction.ts
```

9eba1706 feat(operations): bulk operations by filter, search by request data, job dashboard (CG-O01, CG-O04, CG-O05) - S142

```
M backend/app/src/main/java/com/orqix/runtime/internal/BulkOperationWorker.java
M backend/app/src/main/java/com/orqix/runtime/internal/BulkOperations.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunQueries.java
M backend/app/src/test/java/com/orqix/runtime/internal/BulkAndSearchTest.java
M contracts/openapi.yaml
M docs/admin-guide.md
A docs/build/S142-bulk-search-job-dashboard.md
M docs/user-guide.md
M frontend/src/api/schema.d.ts
A frontend/src/features/runs/BulkActionsPanel.test.tsx
A frontend/src/features/settings/JobDashboardPanel.test.tsx
M frontend/src/features/studio/inspector/FieldListEditor.tsx
M frontend/src/features/studio/model/ops.ts
A frontend/src/features/studio/model/searchable.test.ts
M frontend/src/features/studio/model/types.ts
```

697ba95c fix(studio): collapsed palette keeps full-size buttons and scrolls

```
M frontend/src/features/studio/Palette.tsx
```

ef2b6ef3 feat(analytics): S145 analytics read model, heatmaps, KPI targets and human work (CG-N01, CG-N03, CG-N05)

```
A backend/app/src/main/java/com/orqix/analytics/KpiAlertHook.java
A backend/app/src/main/java/com/orqix/analytics/api/AnalyticsController.java
A backend/app/src/main/java/com/orqix/analytics/api/AnalyticsDtos.java
A backend/app/src/main/java/com/orqix/analytics/internal/AnalyticsQueries.java
A backend/app/src/main/java/com/orqix/analytics/internal/AnalyticsService.java
A backend/app/src/main/java/com/orqix/analytics/internal/KpiEvaluator.java
A backend/app/src/main/java/com/orqix/analytics/package-info.java
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/runtime/internal/AnalyticsOutboxWriter.java
A backend/app/src/main/java/com/orqix/runtime/internal/AnalyticsProjector.java
A backend/app/src/main/java/com/orqix/runtime/internal/AnalyticsTaskTimes.java
M backend/app/src/main/resources/application-test.yaml
A backend/app/src/main/resources/db/migration/V380__analytics_read_model.sql
A backend/app/src/test/java/com/orqix/analytics/AnalyticsApiTest.java
A backend/app/src/test/java/com/orqix/runtime/internal/AnalyticsTaskTimesTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M contracts/openapi.yaml
A docs/build/S145-analytics-read-model.md
M frontend/src/api/schema.d.ts
A frontend/src/features/analytics/HeatmapOverlay.tsx
A frontend/src/features/analytics/HumanWorkCard.tsx
A frontend/src/features/analytics/KpiOverviewCard.tsx
A frontend/src/features/analytics/StudioHeatmap.tsx
A frontend/src/features/analytics/analytics.test.tsx
A frontend/src/features/analytics/api.ts
A frontend/src/features/analytics/heatmap.ts
A frontend/src/features/analytics/useStudioHeat.ts
M frontend/src/features/insights/InsightsPage.tsx
M frontend/src/features/runs/diagram/RunDiagram.tsx
M frontend/src/features/runs/diagram/RunEdge.tsx
M frontend/src/features/studio/CanvasView.tsx
M frontend/src/features/studio/RouteEdge.tsx
A frontend/src/mocks/analytics.ts
M frontend/src/mocks/insights.ts
```

58e651de fix(analytics): mount the Studio heatmap only for run.read so canvases without a query client still render

```
M frontend/src/features/analytics/StudioHeatmap.tsx
M frontend/src/features/analytics/useStudioHeat.ts
M frontend/src/features/studio/CanvasView.tsx
```

1d1937c3 feat(decisions): Orqix expressions as decision table entries (S146 part A, CG-D04)

```
M backend/app/src/main/java/com/orqix/cases/CaseService.java
M backend/app/src/main/java/com/orqix/catalog/api/CatalogDtos.java
M backend/app/src/main/java/com/orqix/catalog/internal/DecisionCsv.java
M backend/app/src/main/java/com/orqix/catalog/internal/DecisionDmn.java
M backend/app/src/main/java/com/orqix/catalog/internal/DmnXml.java
M backend/app/src/main/java/com/orqix/catalog/internal/WorkflowService.java
M backend/app/src/main/java/com/orqix/integration/agenttools/AgentDecisionTools.java
M backend/app/src/main/java/com/orqix/runtime/engine/DecisionTaskDelegate.java
M backend/app/src/test/java/com/orqix/api/DecisionCsvApiTest.java
M backend/app/src/test/java/com/orqix/catalog/internal/DecisionCsvTest.java
M backend/app/src/test/java/com/orqix/catalog/internal/DecisionDmnTest.java
A backend/app/src/test/java/com/orqix/runtime/DecisionExpressionsRuntimeTest.java
M backend/compiler/src/main/java/com/orqix/compiler/DecisionEvaluator.java
M backend/compiler/src/main/java/com/orqix/compiler/DecisionTables.java
M backend/compiler/src/main/java/com/orqix/compiler/WorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/expr/Expression.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/DefaultWorkflowCompiler.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/bpmn/BpmnGenerator.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/dmn/DmnExport.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/dmn/DmnGenerator.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/dmn/ExpressionDmn.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/eval/DefaultDecisionEvaluator.java
A backend/compiler/src/main/java/com/orqix/compiler/internal/eval/ExpressionTable.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/ModelReader.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/model/WorkflowDefinition.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/BindingRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/ExpressionRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/FieldTypes.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/LogicalConditions.java
```

```
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/Tier1Rules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/rules/TypeRules.java
M backend/compiler/src/main/java/com/orqix/compiler/internal/sim/SimulationRun.java
A backend/compiler/src/test/java/com/orqix/compiler/DecisionExpressionCasesTest.java
A backend/compiler/src/test/java/com/orqix/compiler/DecisionExpressionsCompilerTest.java
M contracts/openapi.yaml
M contracts/workflow-v0.schema.json
A docs/build/S146-parts/A-decision-expressions.md
A examples/fixtures/decision-expression-cases.json
M frontend/src/api/schema.d.ts
M frontend/src/features/runs/DecisionEvidence.test.tsx
M frontend/src/features/runs/DecisionEvidence.tsx
M frontend/src/features/studio/inspector/ConditionEditor.tsx
A frontend/src/features/studio/inspector/DecisionFormulaCells.tsx
M frontend/src/features/studio/inspector/DecisionInspector.test.tsx
M frontend/src/features/studio/inspector/DecisionInspector.tsx
M frontend/src/features/studio/model/decisionLogic.ts
M frontend/src/features/studio/model/expressionScope.ts
M frontend/src/features/studio/model/fieldRefs.ts
M frontend/src/features/studio/model/summary.ts
M frontend/src/features/studio/model/types.ts
M frontend/src/features/studio/test/DecisionGrid.tsx
M frontend/src/features/studio/test/Playback.tsx
M frontend/src/features/studio/test/caseGenerator.ts
M frontend/src/features/workflows/diff.ts
A frontend/src/lib/expr/decision.test.ts
A frontend/src/lib/expr/decision.ts
M frontend/src/mocks/decisionCsv.test.ts
M frontend/src/mocks/decisionCsv.ts
M frontend/src/mocks/decisionDmn.test.ts
M frontend/src/mocks/decisionDmn.ts
A frontend/src/mocks/decisionExpressions.test.ts
M frontend/src/mocks/decisionPolicy.test.ts
M frontend/src/mocks/engine.ts
M frontend/src/mocks/handlers.ts
M frontend/src/mocks/simulator.test.ts
M frontend/src/mocks/simulator.ts
M frontend/src/mocks/validate.ts
```

e7f45048 docs(S144): full verify totals

```
M docs/build/S144-alerts-retention-archiving.md
```

4c2c6273 feat(runtime): S143 mapped release migration, migration batches and guarded moves (CG-O02, CG-O03)

```
M backend/app/src/main/java/com/orqix/audit/AuditWriter.java
A backend/app/src/main/java/com/orqix/runtime/api/MigrationAndMoveController.java
A backend/app/src/main/java/com/orqix/runtime/api/MigrationMoveDtos.java
M backend/app/src/main/java/com/orqix/runtime/engine/EngineCommands.java
M backend/app/src/main/java/com/orqix/runtime/evidencepack/RunEvidencePack.java
A backend/app/src/main/java/com/orqix/runtime/internal/MappedMigrations.java
A backend/app/src/main/java/com/orqix/runtime/internal/MigrationBatches.java
A backend/app/src/main/java/com/orqix/runtime/internal/MigrationRules.java
A backend/app/src/main/java/com/orqix/runtime/internal/MoveRules.java
A backend/app/src/main/java/com/orqix/runtime/internal/RunFacts.java
M backend/app/src/main/java/com/orqix/runtime/internal/RunMigrations.java
A backend/app/src/main/java/com/orqix/runtime/internal/RunMoves.java
A backend/app/src/main/java/com/orqix/runtime/internal/StepGraph.java
M backend/app/src/main/java/com/orqix/runtime/internal/TaskStore.java
M backend/app/src/main/java/com/orqix/shared/problem/ProblemCode.java
A backend/app/src/main/resources/db/migration/V360__run_migration_batches_and_moves.sql
A backend/app/src/test/java/com/orqix/runtime/MigrationAndMoveRulesTest.java
A backend/app/src/test/java/com/orqix/runtime/RunMigrationWithMappingTest.java
A backend/app/src/test/java/com/orqix/runtime/RunMoveTest.java
M backend/app/src/test/java/com/orqix/security/SecurityMatrixTest.java
M backend/app/src/test/java/com/orqix/support/TestUsers.java
M contracts/openapi.yaml
M docs/admin-guide.md
A docs/build/S143-instance-migration-and-move.md
M frontend/src/api/endpoints.ts
M frontend/src/api/schema.d.ts
A frontend/src/features/runs/MigrationAndMoveDialogs.test.tsx
A frontend/src/features/runs/MigrationAndMoveDialogs.tsx
M frontend/src/features/runs/RunDetailPage.tsx
```

```
M frontend/src/lib/events.ts
M frontend/src/lib/problemMessages.ts
M tools/contract-check/contract_check.py
```

64dcd256 docs(build): S142 results

```
M docs/build/S142-bulk-search-job-dashboard.md
```

5a551314 docs(analytics): S145 verification results

```
M docs/build/S145-analytics-read-model.md
```

86bd0d69 test(explain): pass the S142 attention errorCode in RunFactsTest after the Wave D merge

```
M backend/app/src/test/java/com/orqix/runtime/explain/RunFactsTest.java
```

c1956b19 docs(S146): build note for decision formulas, document extraction, operator explain and AI skills pages

```
A docs/build/S146-completion-expressions-extraction-ai-pages.md
D docs/build/S146-parts/A-decision-expressions.md
D docs/build/S146-parts/B-document-extraction.md
D docs/build/S146-parts/C-operator-explain.md
D docs/build/S146-parts/D-ai-skills-studio.md
M frontend/src/features/studio/model/notation.test.ts
```

fa0319cd fix(retention): purge the request-linked tables added by other Wave D slices (V401)

```
A backend/app/src/main/resources/db/migration/V401__purge_wave_d_tables.sql
M docs/build/S144-alerts-retention-archiving.md
```

7e63a8ed test(cases): count stage-overdue recipients instead of assuming two operators

```
M backend/app/src/test/java/com/orqix/runtime/CaseStagesTest.java
```